



Sardar Patel Institute of Technology
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-
400058, India
(Autonomous College Affiliated to University of Mumbai)

1.3.2 – Institutional Programme Notices with course modules.

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Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai – 400 058-India.
(Autonomous Institute Affiliated to Mumbai University)

NOTICE

Date: 15/01/2024

All the students of Second Year (UG), Third Year (UG) Final year (UG), First Year (PG) who have taken NPTEL are hereby reminded that NPTEL courses Enrolment in many courses is about to close. You are required to complete the courses enrollment at the earliest. After that you must register for the course (examination) by paying requisite fees online at the NPTEL website (Retain the fees receipt, useful to claim the refund) While registering select the checkbox to " Can we share your scores with your college *" as Yes. This will help in verification of your course result. Failing to which your results may get delayed.

After registering to the course please update to Dr. Sukanaya Kulkarni (Controller of Examinations and

Dr. Y. S. Rao

Dean Academics & Research, SPIT Mumbai





Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai – 400 058-India.
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NOTICE

Date: 05/01/2022

All the students of Second Year (UG), Third Year (UG) Final year (UG), First Year (PG) who have taken NPTEL are hereby reminded that NPTEL courses Enrolment in many courses is about to close. You are required to complete the courses enrollment at the earliest. After that you must register for the course (examination) by paying requisite fees online at the NPTEL website (Retain the fees receipt, useful to claim the refund) While registering select the checkbox to " Can we share your scores with your college *" as Yes. This will help in verification of your course result. Failing to which your results may get delayed.

After registering to the course please update to Prof. Govind Haldankar (Controller of Examinations

Dr. R. G. Sutar

Dean Academics, SPIT Mumbai





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Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai – 400 058-India.

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NOTICE

Date: 05/09/2020

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Dr. R. G. Sutar

Dean Academics, SPIT Mumbai



A handwritten signature in blue ink, appearing to read 'G. T. Haldankar'.



Bharatiya Vidya Bhavan's

Sardar Patel Institute of Technology

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NOTICE

Date: 20/04/2020

All the students of Second Year (UG), Final year (UG) students, First Year (PG) who have taken NPTEL/MOOC are hereby informed that their grade calculation will be based on the assessment of the assignments completed in the NPTEL /MOOC course.

Dr. R. G. Sutar

Dean Academics, SPIT Mumbai





Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics Engineering](#) > Advanced Instrumentation

Advanced Instrumentation

- **ELSD43: LabVIEW Programming-I**
- ELSD44: Signal Processing with LabVIEW
- ELSD45 : Image Processing with LabVIEW
- ELSD46: Data Acquisition and Control-II
- ELSD47: PLC and SCADA Programming
- ELSD48: Development of Healthcare Platform



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics Engineering](#) > [Advanced Instrumentation](#) > ELSD43: LABVIEW PROGRAMMING – I

ELSD43: LABVIEW PROGRAMMING – I

[Learners Login](#)

About the Course

Academic research and development usually encompasses discovery, innovation, experimentation, and creation; however, in today's highly competitive and global economy, it also involves patents, licensing, technology transfer, and partnerships with industry. Virtual instrumentation is the combination of user-defined software and modular hardware that

implements custom systems (“virtual instruments”) with components for acquisition, processing/analysis and presentation.

National Instruments introduced the concept of virtual instrumentation more than 25 years ago and now offers an extensive platform of hardware and software for creating virtual instruments. Since its inception, the virtual instrumentation approach has gained widespread acceptance around the world. For instance, in 2004 National Instruments sold more than 6 million channels of virtual instrumentation in 90 countries.

This training will be an intensive 40 hours course that will involve the building, Simulating and Testing VI's with National Instruments Labview Software. This Course will teach learners deep understanding of Graphical programming language and its capabilities to quickly prototype system from various domain thereby greatly reducing time to market.

Pre-requisites

Basic Understanding of any Programming Language Constructs

Industries where this course will be useful

In India, the main users of LabVIEW are

1. Alliance partners of National Instruments like Captronic Systems ,Soliton, Apna etc.
2. Govt. Research establishments.
3. MNC Software service companies like HCL, Wipro, Infosys etc.
4. Product companies (GE, Nokia,Moog etc)

Course Outcomes

1. Ability to model Complex system
2. Ability to use Graphical Programming Language
3. Ability to Simulate and Test behavior of System
4. Ability to use various Hardware interface with Graphical Programming Language
5. Ability to improve Programming Skills
6. Improving Capabilities to build Portable platform independent software



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Course Layout

A) Labview Environment

- B) Labview Data Types
- C) Graphs and Charts
- D) Debugging Tools
- E) Loops
- F) Structures
- G) Arrays
- H) Cluster
- I) Introduction to SubVI's
- J) File I/o



S. Chandhraj



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > [Cyber-Physical and Embedded Systems](#) > ETSD13: PCB Design

ETSD13: PCB Design

About the course

An important concept in electronics is the system design using printed circuit board (PCB). In this course Basic concepts such as terminology, design process and assembly methods of PCB will be discussed. Points like multiple layer of PCB, designing for varied applications such as high frequency, low power, power electronics etc. will be covered.

In depth discussion on minimum system design, EDA design tools, schematic design, significance of various libraries and footprints. Demonstration will be using “Eagle” software. New component/library design and gerber file creation will be taught.

Soldering is a technique in which two or more metal items are joined together by melting and flowing a filler metal (solder) into the joint, the filler metal having a lower melting point than the adjoining metal. Soldering differs from welding in that soldering does not involve melting the work pieces. As a part of the course hands on session will be conducted on “soldering” and “de-soldering” techniques for mounting various electronic components which includes Surface Mounted Devices (SMD).



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Pre-requisites:

No

Course Outcomes:

1. able to design circuit diagrams and multilayer PCBs
2. able to design New components and Libraries
3. able to solder and de-solder various types of electronic components

Course Material:

1. Why Digital **(PPT)**
2. PCB Layout Tips **(PPT)**
3. Cadsoft Eagle **(PPT)**

Course Videos:

1. PCB Design Rules:
<https://www.youtube.com/watch?v=yx2l1EFZp-c>
2. PCB Design using Cadsoft Eagle:
<https://www.youtube.com/watch?v=MRZl6FVzZRc>
3. Soldering Practice
<https://www.youtube.com/watch?v=RV1WSldLJyY>

Course Instructors:

Dr. Y S Rao

Prof. G T Haldankar



G T Haldankar



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > [Cyber-Physical and Embedded Systems](#) > ETSD14: Internet of Things (IoT)

ETSD14: Internet of Things (IoT)

About the course

Mobile and wireless communications applications have a clear impact on improving the humanity wellbeing. From cell phones to wireless internet to home and office devices, most of the applications are converted from wired into wireless communication. Smart and advanced wireless communication environments represent the future technology and evolutionary development step in homes, hospitals, industrial, vehicular and transportation systems.

In this course various wireless communication devices will be covered such as Wi-Fi, Bluetooth, SIM808 GPS/GSM/GPRS, Zigbee. System design, programming and implementation of various protocols will be implemented using “AT” commands. As a part of the course students will be to practice and implement Web services and API, examples like:

- To setup Wi-Fi device as Server and send the data to the Mobile web browser
- Wi-Fi device as Client to get Real-time data from a website and display on Client (PC Terminal)
- To setup Wi-Fi device as client send the data to the website for graphical plotting
- To Get Real Time Weather Data from a Website and Display on Browser

(Mobile)



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Pre-requisites:

No

Course Outcomes:

1. able to design circuit diagrams for Wi-Fi, Bluetooth, SIM808 GPS/GSM/GPRS, Zigbee.
2. able to program communication devices using PC and “AT” Commands
3. able to implement IoT applications using PC, Mobile, Web services, API, etc.

Course Material:

1. [Wireless Communication](#)
2. [AT Commands](#)
3. [IoT LAB-1](#)
4. [IoT LAB-2](#)
5. [IoT LAB-3](#)
6. [IoT LAB-4](#)

Course Videos:

<https://www.youtube.com/user/EMTRONTechnologies>

1. Wireless Communication Board Design Part-1
2. Wireless Communication Board Design Part-2
3. IoT Lab-1 (To set up server and send data over the web browser)
4. IoT Lab-2 (To get real time data from a website)
5. IoT Lab-3 (To set up server and send data over the Thingspeak)
6. IoT Lab-4 (To Get Real Time Weather Data from a Website and Display on Browser (Mobile))
7. IoT Lab-5 (PC to PC communication using Wi-Fi)

Course Instructor:

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Dr. Y S Rao
Prof. G T Haldankar



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > [Simulation and Modeling for Wireless and Mobile Communication](#) > ETSD23: Programming in Scilab/MATLAB-I

ETSD23: Programming in Scilab/MATLAB-I

Course Co-ordinator:

Prof. Sukanya A Kulkarni
Associate Professor
EXTC Dept., S.P.I.T
Email: sukanya_kulkarni@spit.ac.in

Schedule: 8th – 23rd Sept. 2017 (Friday 4pm -6pm and Saturday 10am-5 pm)

Contact for Enrolment

Prof. Sukanya A Kulkarni
sukanya_kulkarni@spit.ac.in

Prof. Lynette Tuscano
lynette.tuscano@spit.ac.in



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The course, intended for students with no programming experience, provides the foundations of programming in MATLAB®. Variables, arrays, conditional statements, loops, functions, and plots are explained. Functions related to Linear Algebra and Signal Processing tool box will be covered. At the end of the course, students should be able to use MATLAB in their own work.

The course mostly follows the official MATLAB Manual, available from The MathWorks.

The Course Structure will be of 40 hours including pre-lab session, hands on and mini project. This course will teach the learners how to use MATLAB efficiently.

Course Outcomes

Student will be able to

1. Develop programs in MATLAB
2. Evaluate, analyze and plot results.
3. Perform mathematical Modelling in MATLAB
4. Good understanding of Linear algebra and Signal processing concepts

Course Instructor

Prof. Sukanya A kulkarni

Prof. Lynette Tuscano

Course Layout

10 hours self learning, 20 hrs Hands on session(1 session=2 hours) and 10 hours miniproject.

- Session 1 : Basic commands ,Expressions, Indexing
- Session 2 : Matrices manipulation-I
- Session 3 : Matrices manipulation-II
- Session 4 Data Analysis
- Session 5 : Graphics and plotting
- Session 6 :Polynomials and Symbolic functions
- Session 7 : Numerical Methods
- Session 8: Calculus
- Session 9 : SignalProcessing Toolbox
- Session 10 : Linear algebra using MATLAB



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > [Networking and Security](#) > Linux System Administration

Linux System Administration

Why Learn Linux?

- Internet, Super Computers, Smart Devices, Space Technologies, Corporation's , Credit Card Systems, Sensor Devices , Submarine, Watches , Networking , Raspberry Pi, and over a billion Smartphones is powered by Linux.
- With rise of IoT(Internet of Things) , Linux is definitely a mandatory skill needed to survive in Technology domain in coming years
- Recruiters prefer ease to work in Linux environment as an essential skill*over other skills in the domain. Linux background provides tremendous *job opportunities
- Top tier companies such as Google, Facebook, IBM, Oracle and many other work with Linux or either of its UNIX based derivatives
- Major Foreign Universities , IITs and IISc widely use Linux environments for Desktop Computing and consider Linux knowledge as key prerequisite for the course

Highlights:

- 100% Hands-On Experience Courses
- Certificate Course
- Additional lab exercises
- Individualized learning experience that best fits with your current background and skill level. (One-Student-One-Computer)

Course Content:

Course covers Linux OS Essentials, System Administration, Network Administration,



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Server and Security Administration, Troubleshooting & Other nitty gritty of working in Linux Environment . Exposure to Popular GNU/Linux Distributions will be given such as Debian, Ubuntu, Red Hat, Fedora , Slacko, etc

Course Layout: Session:1-18

1. Linux Commands: Work Effectively on the Unix Command Line
2. Perform Basic File Management
3. Create Partitions and Filesystems
4. Control Filesystem Mounting and Unmounting
5. Installation of Linux, Multiboot environment
6. Linux Booting, boot loader management (Grub, lilo etc)
7. User administration
8. Software Management (Repository Server)
9. Backup and restore (Tar, ISO methods)
10. Automate and scripting (Shell Scripting, scheduling, rsync etc)
11. Process Management
12. Memory management
13. Kernel compilation
14. Network Administration
15. Server Administration (Apache Web server, FTP, SSH, NFS, DNS, Sendmail etc)
16. Security Administration (Firewall, IDS, OS Hardening)
17. Network Troubleshooting using tcpdump and Wireshark
18. Rescue environment

Who Should attend:

1. Engineering students and staff.
2. System/Network Administrators



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3. Network Analyst/Manager
4. System/Web Developers

About the Instructors:



Prof Dayanand Ambawade has over 20 years of experience in Linux & Networking and has been author of two popular books, namely, Linux Lab and Advanced Computer Networks. He is a passionate Teacher and known for his discrete methodology of teaching.

You can find his complete profile here: <http://extc.spit.ac.in/faculty/prof-dayanand-ambawade//>



Prof. Manish Parmar has more than 26 years of experience in Microprocessors and Microcontrollers and Computer Architecture.

You can find his complete profile here: <http://extc.spit.ac.in/faculty/prof-manish-parmar/>



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > [ETSD5: Advances in Antenna](#) > ETSD53: Antenna Design, Fabrication and Testing

ETSD53: Antenna Design, Fabrication and Testing

Course Co-ordinator:

Prof. Reena Sonkusare
Associate Professor
EXTC Dept., S.P.I.T
Email: reena_kumbhare@spit.ac.in

Schedule Dates:

29th July 2017 (Saturday) : 4 hrs
5th August 2017 (Saturday): 6 hrs
12th August 2017 (Saturday): 4 hrs
19th August 2017 (Saturday): 6 hrs

Contact for Enrolment

Prof. Reena Sonkusare
reena_kumbhare@spit.ac.in
Prof. Pallavi Malame
pallavi@spit.ac.in



About the course

In the 1890s, there were only a few antennas in the world. These rudimentary devices were primarily a part of experiments that demonstrated the transmission of electromagnetic waves. By World War II, antennas had become so ubiquitous that their use had transformed the lives of the average person via radio and television reception. The number of antennas in the United States was on the order of one per household, representing growth rivaling the auto industry during the same period.

By the early 21st century, thanks in large part to mobile phones, the average person now carries one or more antennas on them wherever they go. This significant rate of growth is not likely to slow, as wireless communication systems become a larger part of everyday life. In addition, the strong growth in RFID devices suggests that the number of antennas in use may increase to one antenna per object in the world (product, container, pet, banana, toy, cd, etc.). This number would dwarf the number of antennas in use today. Hence, learning a little (or a large amount) about antennas couldn't hurt, and will contribute to one's overall understanding of the modern world.

Major Classification of Antenna Includes:

1. Yagi-Uda Antenna
2. Horn antenna
3. Antenna array
4. Parabolic Reflectors and
5. Patch Antenna : Which includes PIFA the Heart and Soul of any Mobile Device.

We are pleased to come up with the 40hrs Skill Development Training Program on “Antenna Design, Fabrication and Testing” which deals with the basic understanding, designing, simulation, Fabrication and Testing of variety of Antennas.

Pre-requisites:

Basic Physics

Industries where this course will be useful

ISRO, SAMEER, BEL, CDAC, DRDO, Rosenberger India, Keysights (Agilent) and many more small scale Industries. Remember without Antenna there cannot be communication!!

Course Outcomes

1. Ability to understand and design antennas according to Requirement and applications
2. Ability to design and Fabricate different antennas based on feeding type



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3. Ability to design Planar Antenna
4. Ability to troubleshoot and test antennas based on parameters evaluated.

Course Instructor:

Prof. Reena Sonkusare (Pursuing PhD)

Prof. Pallavi Malame

Course Layout

10 hours self learning, 20 hrs Hands on session(1 session=2 hours) and 10 hours miniproject.

Session 1: CAD Simulation of Probe Feed Rectangular Patch Antenna using Constant Parameters

Session 2: CAD Simulation of Probe Feed Rectangular Patch Antenna using Variable Parameters

Session 3: CAD Simulation of Inset Feed Rectangular Patch Antenna using Constant Parameters

Session 4: CAD Simulation of Inset Feed Rectangular Patch Antenna using Variable Parameters

Session 5: IEEE Paper Implementation using CAD Simulator

Session 6: Graded Assignment I: IEEE Paper Implementation using CAD Simulator

Session 7: Graded Assignment II: IEEE Paper Implementation using CAD Simulator

Session 8: Fabrication and Testing of Antenna using VNA I

Session 9: Fabrication and Testing of Antenna using VNA II

Session 10: Session on Technical Paper Writing and Evaluation



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Computer Engineering](#) > [Expert System Design and Development](#) > CESD13 : Expert System Tools

CESD13 : Expert System Tools

Course Layout

CESD13 : Expert System Tools

The first module is on **Expert System Tools**, In this, We will cover the basics of Expert systems which will include fundamental knowledge. This is required to understand the Rule based and knowledge based expert system and its constituents, with hypothesis testing and rule derivation. Applied mathematical concepts will be taught, to understand designing Expert systems. The course will be accompanied by hands-on problem solving, with programming in Java. It also covers the tools like NNEasy, Joone and other important open source tools available.

Course Instructor #1



Dhananjay

Dhananjay Ramrao Kalbande is currently a Professor and Head of Computer Engineering and Dean(Industry Relations), at Sardar Patel Institute of Technology, Andheri (West), Mumbai, India. He has completed **B.E. in Computer Technology** from Nagpur University in 1997 and **Master of Engineering in Information Technology** in May 2005, from Vivekanand Education Society's Institute of Technology(VESIT),

Mumbai University, Mumbai, India. He has obtained **Ph.D** in Technology from University of Mumbai, Mumbai in 2011. He has been awarded as a **Post-Doctorate** from Tata Institute of Social Sciences(T.I.S.S.) in 2016.He was also associated with T.I.S.S. as a Senior Research Fellow on the NCW-TISS Project, funded by National Commission for Women, Govt.of India and Research Fellow by Tata Trust & M.I.T.(U.S.A.) on the CLix Project. He has over 17+ Years of experience in teaching & research. His research interests include Soft computing (Neural Networks, Fuzzy Logic),Computer Network, Human Machine Interaction Decision making, Mobile application development for social cause, ICT for semi rural development for social cause. He has authored 4 books, namely Graphical User Interface (Pareen Publications), MIS (Pareen Publications), Human Machine Interaction(Wiley Publications), Digital Forensic(Wiley Publications) for Engineering Graduate students. He has delivered and conducted workshops and seminar talks on NS2,Neural Network & Fuzzy Logic ,VB.Net and ADO.Net. He was awarded as distinguished professor & distinguished HOD by CSI in 2017.

Course Instructor #2



Ananya Jain is currently working at **JPMorgan Chase** as an application developer primarily working with Java and Web technologies. He obtained B.E. in Computer Engineering from University of Mumbai in 2015. He worked on numerous projects during his engineering course including computer engineering department website and handled consultancy project of website development. He is passionate about technology and has keen interest in the field of Soft Computing and Human-Computer Interaction.

Course Instructor #3



Behaadhari



Sunny Shah is currently working as **Senior Software Developer at Morgan Stanley**. He has completed his Bachelors in Computer Science from Sardar Patel Institute of Technology, University of Mumbai in 2015. Apart from the experience working on high frequency trading applications in Morgan Stanley, he has immense experience in competitive programming. He also co-founded “CannyBrain.com”, which started as an educational initiative and gradually became commercialized, He worked on many other projects including Computer Department website for SPIT and Lunchex.co web application. He has great interest in domains of Machine Learning, AI and Distributed Application development.

Session – 1 : Rule Based Expert System

Session – 2 : Knowledge based expert systems

Session – 3 : Types of an Expert System

Session – 4 : Hypothesis Testing

Session – 5 : Rule derivation



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Computer Engineering](#) > [Expert System Design and Development](#) > CESD14: Application development using Neural Network and Fuzzy Logic

CESD14: Application development using Neural Network and Fuzzy Logic

Course Layout

In this, we will cover the basics of learning with demonstration by using perceptron Training algorithm. The gradient descent algorithm in Backpropagation helps participants to understand and apply to build supervised expert system for any real-life problems.

It covers the use of the Gaussian method in Radial Basis function with example.

This module covers the design of unsupervised system using Hopfield network.

The application like signature verification, pattern recognition can be design using associative memory(BAM).This module focusing on handling uncertainty , vagueness issues while designing an expert system using Fuzzy logic. The course will be accompanied by hands-on problem solving with programming in Java and Python and also covers the tools like ,OpenNN, Joone, FuzzyTech and other important open source tools available.**Contents**

Perceptron, gradient descent algorithm, BPN, RBF Networks, Hopfield Network, BAM

OpenNN, Java, JooNE



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Fuzzy Rules and Fuzzy Set Operations, First Order Logic, Fuzzy Predicate Logic
fuzzyTECH , Java

Course Instructor #1



Dhananjay Ramrao Kalbande is currently a Professor and Head, Computer Engineering and Dean(Industry Relations), Sardar Patel Institute of Technology, Andheri (West), Mumbai, India. He has completed B.E. in Computer Technology from Nagpur University in 1997 and Master of Engineering in Information Technology in May 2005, from Vivekanand Education Society's Institute of Technology(VESIT), Mumbai University, Mumbai, India. He has obtained **Ph.D** in Technology from University of Mumbai, Mumbai in 2011. He has been awarded as a **Post-Doctorate** from Tata Institute of Social Sciences(T.I.S.S.) in 2016. He was also associated with T.I.S.S. as a **Senior Research Fellow** on the NCW-TISS Project, funded by National Commission for Women, Govt. of India from 23rd July 2016-10th Oct. 2016 and **Research Fellow by Tata Trust & M.I.T.(U.S.A.)** on the CLix Project from 20th Feb to 19th May 2017. He is having over 17+ Years of experience in teaching & research. His research interests include Soft computing (Neural Networks, Fuzzy Logic), Computer Network, Human Machine Interaction Decision making, Mobile application development for social cause, ICT for semi rural development for social cause. He has authored the 4 books namely Graphical User Interface(Pareen Publications), MIS(Pareen Publications), Human Machine Interaction(Wiley Publications), Digital Forensic(Wiley Publications). He has delivered and conducted various workshops on Neural Network & Fuzzy Logic ,VB.Net and ADO.Net. He is also awarded as distinguished professor & distinguished HOD by CSI in 2017. Currently working with Research Project titled "Analytical Study of Mosquito Borne Diseases: Analytical Tool development " with Centre for Excellence in Basic Science University of Mumbai-Dept of Atomic Energy, Mumbai since 29/06/2017.

Course Instructor #2



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Sunny Shah is currently working as **Senior Software Developer at Morgan Stanley**. He has completed his Bachelors in Computer Science from Sardar Patel Institute of Technology, University of Mumbai in 2015. Apart from the experience working on high frequency trading applications in Morgan Stanley, he has immense experience in competitive programming. He also co-founded “CannyBrain.com”, which started as an educational initiative and gradually became commercialized, He worked on many other projects including Computer Department website for SPIT and Lunchex.co web application. He has great interest in domains of Machine Learning, AI and Distributed Application development.

Course Instructor #3



Mohsin Karovaliya is currently working as a Software Developer at **BookMyShow**. He has completed Bachelors in Computer Engineering from Sardar Patel Institute of Technology in 2015. He has worked across different domains and teams involving Nodejs Applications, IOS development and Big Data. He likes developing highly-scalable and real-time systems. He recently developed a Chat Messaging system from the group up for BMS. He is currently working on Big Data technologies which involves developing Audience Intelligence system

Scheduled Dates for Batch : Third week of Jan.17
(Actual dates will be displayed soon)



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Computer Engineering](#) > [CGI Scripting and Text Processing](#) > CESD23: Shell Scripting

CESD23: Shell Scripting

Course Duration:

Part I & II – Sept 04, 2017 – Sept 28, 2017

Part III – Oct 03, 2017 – Oct 13, 2017

Course Fee:

Rs. 5000/- (Rs.3000/- for S.P.I.T. students)

Last date for enrollment: 07 August, 2017



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About the course

UNIX and Linux operating systems have traditionally been the platform of choice for database, big data analysis and financial applications because of the rapid growth of excellent development tools. These tools can be used for the installation of software, software development, monitoring of system resources, and management of systems through Shell scripting. Shell scripting can automate and allow the fast development of software technologies on Java EE Server like IBM Websphere, Oracle WebLogic, Apache Tomcat, Oracle RDBM etc on UNIX and Linux Environment. CGI along with Shell scripting can offer well designed web services for the administration of servers. This course of shell scripting can also help the computer and IT professional from the domain of Database and System Administrators.

This training of Shell scripting is a FIRST course out of the remaining SIX courses in the track of “CGI Scripting and Text Processing”. This training will be an intensive 40 hours which is divided in three parts: I) Basics of Linux Commands and Shell Scripting (10 Hrs), II) Extensive Hands-on-Workshop of Shell Scripting (20 Hrs) and III) Development of Main Project

through Assignments (10 Hrs). This course will teach the learners the extensive Linux commands and Shell scripting.

Pre-requisites

1. Basics of Linux OS
2. Basics of any programming language e.g. C, C++ etc.

Industries where this course will be useful:

Several product-based companies like Google, Amazon and Oracle etc. requires Shell scripting experts for the management, automation and maintenance of high-end servers. Service-based companies like TCS, Infosys, Cognizant, Capgemini, L&T InfoTech may also needs Shell scripting experts.

Course Outcomes

- Able to know the different environment variables of Linux Operating Systems.
- Able to create scripts to automate common tasks in Linux system.
- Able to creating report and online report using CGI and MySQL queries.
- Able to guard against malicious intents against the Linux Operating System.



Course Instructors

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Dr. Anant V. Nimkar received PhD from IIT Kharagpur. Currently, he is working as Associate Professor at Sardar Patel Institute of Technology, Mumbai. He has 15 years of teaching experience. His interest includes Information and System Security, High Performance Computing, Cloud Computing, Graph Theory and Distributed Algorithms. He has proficiency in Linux OS Internals and Administration, Java EE, and Android Programming.

Mr. Jagjit Phull is a computer science engineer by profession who has vast and enriched industry experience of 23 years and currently works as network administrator and developer. He has been working with GNU/Linux and Unix systems from around 1994 and has administered and developed

solutions based on UNIX-AIX/Solaris/HP-UX, GNU/Linux -Debian, Ubuntu, Fedora, CentOS. Also, he automated networking monitoring and presented clustering solutions and distributed clustering. He has hands-on and fluid experience and knowledge in working with different technologies like Hadoop, Cassandra, MongoDB, DCEBIGIP, Netscaler, MySQL, PostgreSQL, Nagios, Cacti, SEC etc.

Prof. Yogesh H. Jadhav is currently working as Assistant Professor in Sardar Patel Institute of Technology, Andheri. He has 5 years of Teaching Experience. He worked as corporate trainer for Livewire India. He worked as Programming Head for Anubhav Institute Pvt. Ltd. He conducted various workshops on Cloud Computing, python, .Net, Linux and Web Development with SEO.

Prof. Abhijeet V. Salunke received his M.E. in computer engineering stream from Mumbai University. Currently he is working as Assistant Professor in Sardar Patel Institute of Technology. He has worked with Cognizant India Pvt. Ltd. as a Programming Analyst in the field of Advance Java along with JSTL technology. He has 5 years of industry and teaching experience. His interest includes Structured and Object Oriented Analysis and Design, Mobile Communication and Computing and Computer Networks, Security and Data analytics. He has proficiency in the domain of Server administration.

Course Layout

Part I – Basics of Linux Commands and Shell Scripting (10 Hrs):

Session – 1:

1. Introduction to Shell Scripting: Sha-Bang, Script Execution, Special Characters, Variables and Parameters.
2. Quoting, Exit and Exit Status, Tests, Operators.

Session – 2:

1. Environment Variables, Loops and Branches.
2. Command Substitution and Arithmetic Expression.

Session – 3:

1. Internal Commands, External Filters, Programs and Commands.



Abhijeet V. Salunke

2. System and Administrative Commands.

Session – 4:

1. Regular Expressions, Here Documents and I/O Redirection.
2. Sub-shells, Restricted Shells, Process Substitution, Functions.

Session – 5:

1. Aliases, List constructs, Arrays, Indirect References and Preliminary CGI Scripting through Bash
2. Network Programming, Debugging, Script Options and Special Directories viz. /dev, /proc, /dev/zero, /dev/null.

Part II – Extensive Hands-on-Workshop of Shell Scripting (20 Hrs):

Session – 1 & 2: Hands on – A high level tool for installing packages, with interaction of the user

Session – 3 & 4: Hands on – Document / Catalog Management systems

Session – 5 & 6: Hands on – Performing basic security check on the OS

Session – 7 & 8: Hands on – Configuration of the applications like apache server in friendly way.

Session – 9 & 10: Hands on – User Administration through CGI – Shell Scripting.

Part III – Development of Main Project through Assignments (10 Hrs):

Each course participant will be assigned a **main course project**. Firstly, the participant has to submit the details of plan of action of 10 hrs. Finally, the participant has to submit **the main course project** for the evaluation and final Course Certificate.



Shandharaj



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Computer Engineering](#) > [Big Data Analytics and Applications](#) > CESD63:
Fundamentals of data analytics

CESD63: Fundamentals of data analytics

About This Specialization

Do you need to understand big data and how it will impact the business? This Specialization is for the professional coming from a background of: Engineering, computer sciences, statistics, mathematics, economy and management. They will gain an understanding of what insights big data can provide through hands-on experience with the tools and systems used by big data scientists and engineers. Previous programming experience is not required! You will be guided through the basics of using Hadoop with MapReduce, Spark, Pig and Hive. By following along with provided code, you will experience how one can perform predictive modeling and leverage graph analytics to model problems. This specialization will prepare you to ask the right questions about data, communicate effectively with data scientists, and do basic exploration of large, complex datasets.

Course Duration:

August 14th, 2017 – September 3rd, 2017

Course Fee: Rs. 5000/- (Rs.3000/- for S.P.I.T. students) Last date for enrollment: 31st July, 2017



Behandharj.

About This Course

Interested in increasing your knowledge of the Big Data landscape? This course is for those new to data science and interested in understanding why the Big Data Era has come to be. It is for those who want to become conversant with the terminology and the core concepts behind big data problems, applications, and systems. It is for those who want to start thinking about how Big Data might be useful in their business or career. It provides an introduction to one of the most common frameworks, Hadoop, that has made big data analysis easier and more accessible — increasing the potential for data to transform our world!

More and more organizations these days use their data a decision supporting tool and to build data intensive products and services. The collection of skills required by organizations to support these functions has been grouped under the term “Data Sciences”. This course will cover the basic concepts of big data, methodologies for analyzing structured and unstructured data with emphasis on the relationship between the Data Scientist and the business needs. The course is intended for the students coming from a background of: Engineering, computer sciences, statistics, mathematics, economy and management.

Data Analytics is the science of analyzing data to convert information to useful knowledge. This knowledge could help us understand our world better, and in many contexts enable us to make better decisions. While this is the broad and grand objective, the last 20 years has seen steeply decreasing costs to gather, store, and process data, creating an even stronger motivation for the use of empirical approaches to problem solving. This course seeks to present you with a wide range of data analytic techniques and is structured around the broad contours of the different types of data analytics, namely, descriptive, inferential, predictive, and prescriptive analytics.

This will be an intensive 40 hour training course that will involve the concepts of big Data analytics with technology. This course will involve one capstone project. A Capstone Project is a larger project designed to help you practice, apply, and showcase the skills you’ve learned.



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This course has a total of six modules, divided into various sub-modules as follows:

- Fundamentals of data analytics
- Prediction analytics: regression and classification
- Prediction analytics: Time series, Decision Tree and Neural Network
- Big Data analytics : Hadoop and SPARK
- Big Data analytics : SCALA
- Data analytics applications

What Background Knowledge is necessary?

This course is for those new to data science. No prior programming experience is needed, although the ability to install applications and utilize a virtual machine is necessary to complete the hands-on assignments.

Industries where this course will be useful

Several Companies in the world are recruiting data scientists and analysts with knowledge and skill of Big Data Technologies.

Course Outcomes

1. To understand algorithmic, theoretical and computational approaches of big data.
2. To understand regression and classification with respective Big data
3. To understand data mining algorithms with respective Big data.
4. To understand HADOOP and SPARK for data analytics.
5. To use SCALA for BIG DATA analytics.
6. To design and develop applications in domains as diverse as genomics, medicine, healthcare, clinical, biological and neuro-informatics, natural robotics, language processing etc.



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Course Creators/Instructors

- Dr. Sudhir N. Dhage, Ph.D.(Technology) in Computer Engineering, VJTI, Mumbai.
Professor, Department of Computer Engineering, SPIT, Mumbai.
- Dr. Lalit, Ph.D.(Technology) in Computer Science and Engineering, IIT(BHU), Varanasi.
Scientist/E, R&D- Computer, Bhabha Atomic Research Centre (BARC), Government of India, Mumbai.
- Prof. Pramod Bide, M.E. Computer Engineering
Assistant Professor, Department of Computer Engineering, SPIT, Mumbai.
- Prof. Yogesh Jadhav, M.E. Computer Engineering
Assistant Professor, Department of Computer Engineering, SPIT, Mumbai.

Course Layout

The first module is on Fundamentals of data analytics, we will cover the basics of Data analytics which will include fundamental of big data concepts. This is necessary to understand and apply on Big Data Applications. The course will be accompanied by hands-on problem solving. This course will help to understand real time problems in Big Data and solve them by using various approaches and methods.

- Session 1: Introduction to Big Data
- Session 2: Clarifications of types of Big Data
- Session 3: Big Data Applications
- Session 4: Issues on Big Data
- Session 5: Random variables and Probability distributions
- Session 6: Regression analysis in statistical analysis of Big Data
- Session 7: Big Data Tools and Technologies



S. Chaudhary

- Session 8: Processing Big Data



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Information Technology](#) > [Game Designing And Deployment](#) > ITSD23 – Introduction to Game Development

ITSD23 – Introduction to Game Development

About the course

Game design is a large field, drawing from the fields of computer science/programming, creative writing, and graphic design. Game designers take the creative lead in imagining and bringing to life video game worlds. Game design is a subset of the field of video game development. Game design is a field with a broad focus.

As such, the skills of a game designer are drawn from the fields of computer science and programming, creative writing and graphic design. Game designers take the creative lead in imagining and bringing to life video game stories, characters, gameplay, rules, interfaces, dialogue and environments. This being the case, a game designer is a cross between a writer, artist and programmer. It is an individual who presents a comprehensive artistic vision, while also possessing the technical skill to oversee and contribute to programming, image rendering, level design, digital editing and other construction aspects of game design.

This training will be an intensive 40 hours course that will involve the design and implementation of various aspects of Game Designing. This course will involve mini-project and one main course project.



S. Chaudhary

Pre-requisites:

Programming Skill

Course Outcomes:

1. Ability to use different Game Engines
2. Ability to program using Unity game engine
3. Ability to design interactive game and deployment in android platform
4. Ability to develop a game using Unity game engine

Course Instructors

Name: Taral Patel
Company Tap Cube Studios
Designation: Co-Founder



Name: Dinesh Iyer
Company: Tap Cube Studios
Designation: Lead Programmer and Designer

Course Layout:

- Session 1: Basic Programming with OOPS
- Session 2: Introduction to Unity Game Engine + Unity Engine with Basics
- Session 3: Vector Mathematics and Physics in Unity



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- Session 4: Introduction to User Interface and Deploying in Android Platform
- Session 5: Self-Study
- Study Different Game Engines
- Analyse a game and fix its flaws and bugs.
- Session 6: Project : Develop a Game

All study materials will be provided in form of links in form of articles and reference videos that we take help of.

Course Co-ordinator #1



Prof. Rupali Sawant
Assistant Professor
IT Dept. SPIT

Course Co-ordinator #2



Prof. Varsha Hole
Assistant Professor
IT Dept. SPIT



Behaudhary



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Information Technology](#) > [Network and IT Infrastructure Security](#) > Network and IT infrastructure

Network and IT infrastructure

About the course

Network devices including firewalls and routers are the gatekeepers to “endpoint” resources and are increasingly using complex software components. These devices are often remotely accessible and whose configuration changes regularly, making them susceptible to vulnerabilities and misconfigurations. To add to this, network and security teams are often separate parts of an organization often leading to incomplete understanding of vulnerable infrastructure.

This training will be an intensive 40 hours course that will involve the design and implementation of various aspects of Network and IT infrastructure. This course will teach the learners simulation and hardware concepts of designing Network and IT infrastructure. This course will involve mini-project and one main course project.

Course Outcomes

1. Ability to describe and analyze the hardware, software, components of a network and the interrelations.
2. Ability to implement Local Area Networks using both static and dynamic addressing techniques including subnetting.
3. Ability to use networking tool to observe and analyze behaviors of networking protocols



Shandhu

Course Instructor:



Mr Mahesh Lawand is currently working as Sr. System Analyst / Presales Consultant at Nihilent Technologies Ltd., Pune, India. He has over 10 years of experience in networking. It spanned Network Presales for Active & Passive network infrastructure, Network Support Services, Remote Technical Support and NOC monitoring-troubleshooting. Expertise in Microsoft Office/Visio, CCW (Cisco Commerce Workspace), Vendor specific configuration tools, Solarwinds / Whatsup-Gold Monitoring Utilities.

His Certifications : CCNA, CCVP.

His Role & Responsibilities :

- To mentor the team and help them build-up the technical solutions and other deliverables.
- Delivering a high quality, competitive techno-commercial networking solutions to the client. Ensure quality and competitiveness of proposals being made and same to be presented to the customer in effective manner
- Analyze customer's business requirement and Put it into technical terms then designing technical solutions to meet their expectations. Assist Account Executives to develop responses to client proposals and requests
- Communicate with clients to understand their current technical environment, key business issues, and future technology requirements. Sizing the components based on specifications and requirement of the clients, follow ups with clients in order to win their confidence & meet their expectations
- Develop and present technical design documents to clients including network analysis, Visio diagrams, configurations, and design validations to ensure clients have the technical tools and information required to be self-sufficient.



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- Ensure timely delivery of all these solutions & different reports to keep the track of all the tasks done
- **Solution Domains** – IP Telephony, LAN, WAN, Wireless, Security & various Unified communication technologies, Structured Cabling.
- **Deliverables:** Bill of Material (BoM) / Statement of Work (SoW) / Response to (RFP, RFI, RFQ) / Network Diagrams (Visio).



Mrs. Renuka Pawar is currently working as Assistant professor in Information Technology department at Sardar Patel Institute of Technology Mumbai. She has over 09 years of experience in teaching.

She has done certification like DCT D-link certified trainer ,CEH -certified ethical hacker v9.0.

She has conducted a workshop on Ethical Hacking in Atharva College of Engineering, Malad. She has also organized and conducted a Two days Hands-on workshop on “Ethical Hacking” at Sardar Patel Institute of Technology Mumbai.

She has assigned a role of Teaching assistant for two week ISTE workshop on computer networking conducted by IIT Bombay.

She has successfully published his research work in Three International conferences and Two Journal papers

Course Layout :

- Session – 1 : Introduction to computer networks, Networking Components and Devices



Renuka Pawar

- Session – 2 : Introduction to TCP/IP Protocol Suite
- Session – 3 : Configure IP subnetting
- Session – 4: Configuration of VLAN and VLAN trunking protocol.
- Session – 5 : Implementing Router on stick
- Session – 6 : Implementation of Static and dynamic routing
- Session – 7 : Configure Default routes, static route
- Session – 8 : Configuration of OSPF ,EIGRP and BGP



Selvaudhraj



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Information Technology](#) > [Banking And Finance](#) > Module 1 – Financial Accounting

Module 1 – Financial Accounting

Overview & Introduction

- Significance of Accounting
- Accounting concepts & conventions and policies
- Double-Entry Book Keeping system

Accounting Cycle

- Journal & Subsidiary Books
- Cash Book & Bank Reconciliation Statements
- Ledger & Trial Balance

Final Accounts

- Preparation of Profit & Loss Statement, Balance Sheet & Cash Flow and use of Software
- Companies Act & SEBI
- IFRS & IND AS

Measurement Concepts



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- Tangible Assets
- Intangible Assets
- Inventory

Latest Trends in Accounting

- Human Resource Accounting
- Inflation Accounting
- Triple Bottom Line concept

Reading of Financial Statements

- Ratio Analysis
- Common Size statements
- Trend Analysis

LEARNING OBJECTIVE OF MODULE:

Student completing the module must know how to use accounting software to write and analyze financial statements, estimate an organization's capability based on financial statements.

Modules will be delivered in an innovative manner, in synchronization with Industry needs, by experts from Industry and maximum use of technology. It will be hands on experience with live case studies and not the traditional model of classroom learning and written examinations. Evaluation will be based on credits gathered by cracking business problems in a creative way and MCQs to verify grasp of concepts.



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Faculty Profile

Dr. Sangeeta Pandit a Chartered Accountant associated with P.D. Kunte & Co. Chartered Accountants, from 1984. Areas of expertise include audit and representative practice, consultancy on accountancy, taxation, finance and strategy. She would represent clients at the income tax office and as counsel before the Commissioner of Income Tax and Income Tax Tribunal.

She is now in academics, head of finance at Sydenham Management Institute. She was also part of Start-up team of Management Institutes- United World School of Business. Malad and MBA program of Bombay Stock Exchange Training Institute. She is a visiting faculty in B Schools of repute in Mumbai and US.

She is an accredited cancer patient counsellor and obtained her PhD in issues relating to healthcare. She works closely with various bodies for financial inclusion. She is actively associated with Institute of Chartered Accountants of India, Bombay Chartered Accountants Society, BSE Brokers Forum and CII IWN. She is an Independent Director in listed companies. She is a sought after Corporate Trainer. She has to her credit research articles and books written on topics related to finance.



S. S. Kulkarni



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Information Technology](#) > [Banking And Finance](#) > Module 2 – Management Accounting

Module 2 – Management Accounting

Overview of Management Accounting

- Planning and control functions
- Functions of Management Accounting
- Financial Accounting and Management Accounting

Cost Classification, Allocation & Methods of Costing

- Unit costing
- Job and Batch costing
- Operating costing

Cost –Volume – Profit analysis

- Break-even point
- Cash break even point
- Operating leverage and risk

Variable Costing and Decision Making

- Meaning and utility of variable costing
- Limitations of variable costing
- Decision making and relevant costs



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Standard costing and variance analysis for cost control

- Development of standards
- Material, Labor and Overhead Standard Costs
- Utility and limitations of Standard Costing
- Significance of Variance analysis
- Computation of variances and analysis

Budgeting for Planning & Control

Emerging Areas

- Target Costing
- Kaizen Costing
- Activity Based Costing



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Information Technology](#) > [Banking And Finance](#) > Module 3 – Understanding Stock Markets & Investments in Securities

Module 3 – Understanding Stock Markets & Investments in Securities

Faculty:

Purv Shah is an independent equity market consultant and a financial markets trainer. His areas of training and content development are mainly Equity and Derivative Markets. He is reputed for delivering hands-on experience of the Capital Markets by role plays, simulations and activities to facilitate learning.

His clients are BSE, FTKMC, MCX, Department of Direct Taxes, Gol, Mphasis, Reliance Money, ICICI Direct etc. He is a reputed Capital Market trainer and mentor.

Sangeeta Pandit is a chartered accountant and has done her doctorate in management. She was in CA practise for more than two decades and has now diversified to academics, training and consultancy. She is closely associated with the activities of ICAI, BCAS, BSE Brokers Forum and CII IWN. She has conducted customized corporate training assignments and is a visiting faculty at US universities. She is an Independent Director in listed companies and is head of department of finance at SIMSREE B-school.



Behanbhay

Financial Planning

- Portfolio Management Process
- Identification of Risks, Measuring and Managing Risk
- Investment Characteristics of Assets, Efficient frontier and Investment's Optimal Portfolio
- Portfolio Planning and Construction

Market Analysis

- Instruments traded in money and capital markets
- Role of derivative securities: systematic risk and other contemporary issues in the financial system.
- Derivative products

Stock & Shares

- Fundamental Analysis
- Technical Analysis
- Efficient Markets
- Stock and Bond Valuation Models
- Futures and options



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Information Technology](#) > [Banking And Finance](#) > Module 4 – Application of statistical tools in finance

Module 4 – Application of statistical tools in finance

- Application of statistical tools in finance
- Fundamentals: Averages (Mean, Median, Range, Mode)
- Measurements: Quartiles, Quintiles, Deciles and Percentiles
- Measures of Dispersion: Variance, Standard Deviation, Semi-variance, Semi-deviation, Chebyshev's Inequality, Coefficient of Variance, Sharpe Ratio
- Symmetry and Skewness in Return Distribution
- Kurtosis in Return Distributions
- Using Geometric and Arithmetic Means
- Linear Models, Time series models
- Customizing plots, Mathematical annotations, RGL:3D visualization in R using OpenGL

Faculty

1. Kavita Laghate
2. Chitra chandrasekhar
3. Sangeeta Pandit



S. Chaudhary



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Information Technology](#) > Quantitative Trading & Financial Analysis

Quantitative Trading & Financial Analysis

About the course

This course will cover the basics on quantitative trading and will give you an overview of how to build rule-based trading strategies. It will also introduce you to different styles used by participants in financial markets. This training will be an intensive 40 hours course that will involve learning about technical analysis, options trading, financial analysis and building your own trading system.

Pre-requisites

Basic knowledge of python and good analytical skills

Industries where this course will be useful

Financial service companies, Derivatives Trading, Quant and Equity Research

Course 1 : Technical Analysis

- What is Technical Analysis and how it is different from Fundamental Analysis
- Classic Chart Patterns and Indicators
- Moving Averages and how to use them to ride the trend ?
- Position Sizing and Money Management Techniques
- Case Study/Games



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Course 2 : Options Trading

- Basics of Options Trading and how are options different from Stocks ?
- Option Greeks and how they affect option pricing
- Option Strategies for different market conditions
- How to protect your portfolio when markets are falling
- Case Studies/Quiz

Course 3 : Building a Trading System

- Trading Basics
- How to create your trading strategy in Python
- Backtesting your trading strategy
- Evaluating strategy performance
- Machine Learning in Trading

Course 4 : How to conduct Financial Research ?

- How to do Financial Analysis of a company
- Understanding Annual Report of a company
- How to tell if a company is cooking it 's Books
- How to do Valuation of a Company and analysing the operating performance
- How to analyse Management Quality
- Final Checklist before Buying Stocks and how to monitor the portfolio
- When to Sell

Course Instructors



Shandhu

Sourabh Sisodiya

Sourabh Sisodiya is the Co-Founder at Quantify Capital, a quantitative research firm with a strong focus on building quantitative investment & trading models to generate stable and consistent alpha.

He has previously worked across Equity research, Derivatives trading and Financial analysis.

He is an alumnus of S.P.I.T and completed his Bachelor of Engineering (B.E.) in Information

Technology. He has completed all the levels of the CFA(Chartered Financial Analyst)

program, most sought after designations for investment professionals. He is also an NSE

Certified Market Professional (NCCMP) from the National Stock Exchange of India Limited.

His team have launched a new fintech venture "Finsense" with a mission to spread Financial

Literacy across the country

<https://www.linkedin.com/in/sourabhsiso>

Hitul Mistry

Hitul Mistry is the Founder of Digiqt Technolabs, a technology consultancy who is helping

clients across the globe in solving their business problems with cutting edge technology.

He has 7+ years of experience in development of large scale enterprise mission-critical and

fault tolerance applications in e-commerce, insurance, finance, medical devices, IOT

domains. He helps startups and corporates in solving their technology engineering

problems.

He has worked with Coverfox Insurance, Kotak Securities and other cutting edge startups.

He is building next generation algorithmic trading platform for Quantify Capital. He has

presented a talk at EuroPython 2015, Bilbao, Spain and PGConf 2018, India.

<https://www.linkedin.com/in/hitulmistry>



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics Engineering](#) > Front End VLSI Design and Verification

Front End VLSI Design and Verification

- ELSD13 : Verilog Programming I
- ELSD14 : Verilog Programming II
- ELSD 15 : Soft Processors and SOC Programming
- ELSD16: Soft Processors and SoC Programming
- ELSD17: Static Timing Analysis
- ELSD18: SystemVerilog Assertions



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics Engineering](#) > Back End VLSI Design

Back End VLSI Design

- ELSD23: Device simulation with TCAD
- ELSD24: SPICE programming
- ELSD25: MOSFET Modeling
- ELSD26: Low Power VLSI
- ELSD27: Mixed Signal VLSI
- ELSD28: Fundamentals of RFIC



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics Engineering](#) > Electronic Product Design

Electronic Product Design

- ELSD33 : Embedded C
- ELSD34 : PCB Design
- ELSD35 : Programming with Silicon Labs Giant Gecko ARM Cortex M3 MCU.
- ELSD36 : Texas Instruments Real Time Operating System.
- ELSD37 : Embedded Course on Texas Instrument TIVA Launchpad by EdGate, Bangalore.
- ELSD38: Androids for Embedded Engineer.
- ELSD39: IOT with TIMSP432 and NodeRED on IBM Cloud



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics Engineering](#) > Advanced Instrumentation

Advanced Instrumentation

- **ELSD43: LabVIEW Programming-I**
- ELSD44: Signal Processing with LabVIEW
- ELSD45 : Image Processing with LabVIEW
- ELSD46: Data Acquisition and Control-II
- ELSD47: PLC and SCADA Programming
- ELSD48: Development of Healthcare Platform



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics Engineering](#) > DSP, Power Electronics and Control

DSP, Power Electronics and Control

- ELSD53: MATLAB Programming
- ELSD54: Solar System Components
- ELSD55: Solar System Design
- ELSD56: Simulation tools for Power Electronics
- ELSD57: Applications of Digital Signal Processors
- ELSD58: Digital Control System



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics Engineering](#) > Antennas and Networking

Antennas and Networking

- ELSD63: Fundamentals of Networking
- ELSD64: Network Simulation Tools
- ELSD65: Simulation of Radiating Elements
- ELSD66: Design, Implementation and Testing of Radiating Elements
- ELSD67: IP Subnetting and Supernating
- ELSD68: Wireless Network Security



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > Cyber-Physical and Embedded Systems

Cyber-Physical and Embedded Systems

- **ETSD13: PCB Design**
- **ETSD14: Internet of Things (IoT) Sensing Systems**
- ETSD15: Internet of Things (IoT) Actuating Systems
- ETSD16: SDR and USRP
- ETSD17: Application of DSP Hardware for Smart Antennas System
- ETSD18: Software Defined Network (SDN) Design



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > Simulation and Modeling for Wireless and Mobile Communication

Simulation and Modeling for Wireless and Mobile Communication

- **ETSD23: Programming in Scilab/MATLAB-I**
- ETSD24: Programming in Scilab/MATLAB-II
- ETSD25: Scripting Languages Perl and Python
- ETSD26: R programming



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- ETSD27: GNU RADIO
- ETSD28: NS2/NS3



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > Networking and Security

Networking and Security

- ETSD33: Fundamentals of Networking
- **ETSD34: Network Operating System and Linux**
- ETSD35: Network Troubleshooting and Switch Configuration
- ETSD36: Firewall Administration
- ETSD37: Server Optimization and Security
- ETSD38: Cryptography and PKI



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > Multimedia Communications Systems

Multimedia Communications Systems

- ETSD43: Fundamentals of Multimedia Standards
- ETSD44: Image and Video Processing using OpenCV and FFMPEG
- ETSD45: LAN Telephony & VoIP Administration
- ETSD46: Video Streaming
- ETSD47: Implementation of CODEC
- ETSD48: Error Correcting Codes



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Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > ETSD5: Advances in Antenna

ETSD5: Advances in Antenna

- [ETSD53: Antenna Design, Fabrication and Testing](#)
- ETSD54: Planar Array Design, Fabrication and Testing
- ETSD55: Wearable Antenna Design, Fabrication and Testing
- ETSD56: UWB Antenna & Mutual Coupling Reduction
- ETSD57: Metamaterials I
- ETSD58: Metamaterials II



Skill Certification for Outcome-based Professional Education

[Home](#) > [Courses](#) > [Electronics and Telecommunication](#) > Telecommunication Engineering Practices

Telecommunication Engineering Practices

- ETSD63: Fundamentals of Reliability Engineering
- ETSD64: Stress Testing Techniques
- ETSD65: Telecommunication Regulations and Safety and Economics
- ETSD66: Wireless Network Planning, Design and Optimization
- ETSD67: Wireless Network Testing
- ETSD68: Technology Obsolescence



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Skill Certification for Outcome-based Professional Education

[Home](#) > Certified Courses on Embedded Software Development from SafeTTy Systems Ltd. (UK)

Certified Courses on Embedded Software Development from SafeTTy Systems Ltd. (UK)



[Course Flyer \[PDF\]](#)

REGISTRATION OPEN

We are Microdevice Technologies, Mumbai. We are the sole exclusive authorized training partners of SafeTTy Systems. We are authorized to conduct training using the technology developed by SafeTTy Systems for which they hold patents in the broader field of the design of safety critical embedded systems using C-programming language.

Brief of SafeTTy Systems Ltd

From their base in the UK Midlands, they provide support for organisations across the world that need to create **real-time** embedded systems that are **safe**, **reliable** and **secure**.

Many of their customers need to achieve compliance with **IEC 61508**, **ISO 26262**, **DO-178** and related international safety standards and guidelines. They help them to meet these requirements through the use of state-of-the-art "[Time-Triggered](#)" (TT) software architectures.

To support their customers, they offer:



R. Chaudhary

- specialised [consultancy](#) services;
- a range of [training courses](#) with an associated [staff certification](#) programme;
- flexible [ReliabilityTTy® Technology Licences](#);
- a range of [DuplicaTTor® Boards](#) and related support packages;
- a series of popular [books](#) on TT systems and functional safety, including the recently-published '[ERES2](#)'.

Who is behind SafeTTy Systems..

The team at SafeTTy Systems Ltd is led by Dr Michael J. Pont CEng.

Michael was awarded a BSc degree (in Electronics and Electrical Engineering) by the University of Glasgow (UK) and a PhD (in Computer Science) by the University of Southampton (UK). He is the author of several successful [books](#) and more than 100 [technical papers](#); he is named inventor / co-inventor on a number of patents and patent applications.

Michael has been working in the field of [time-triggered \(TT\) embedded systems](#) for more than 20 years. He has helped numerous companies to create successful TT designs. Many of these designs have been produced in compliance with international safety standards and guidelines (including IEC 61508, ISO 26262 and DO-178).

Michael began this work on TT systems in the academic sector, latterly as Professor of Embedded Systems Engineering at the University of Leicester (UK). In Leicester, Michael established the Embedded Systems Laboratory, invented the [PTTES “pattern language”](#), and led teams that developed the “NISA” safety agent, and the first single- & multi-core processors that were designed specifically to run TT software. During his time with the University of Leicester, Michael supervised more than 20 successful PhD projects.

In order to commercialise some of the above technology, Michael founded and served as CEO of the University of Leicester “spin out” company TTE Systems Ltd.

Michael currently serves as CEO of SafeTTy Systems Ltd (January 2014 – date).



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What technology is owned by SafeTTy Systems

They have developed technology that is licensed to customers through their [ReliabiliTTy®](#) programme.

This technology includes the following patents:

- Pont, M.J. (2016) “A monitoring unit as well as method for predicting abnormal operation of time-triggered computer systems”, SafeTTy Systems Ltd, GB2526082.
- Pont, M.J. (2016) “A framework as well as method for developing time-triggered computer systems with multiple system modes”, SafeTTy Systems Ltd, GB2526083.

Related international patents have been applied for.

What Trademarks are owned by SafeTTy Systems..

AcquisiTTor, ConnecTTor, CorrelaTTor, DecomposiTTor, DisrupTTor, DuplicaTTor, MoniTTor, PredicTTor, RapidiTty, ReliabiliTTy, SafeTTy, SafeTTy Certified, SafeTTy Systems, TriplicaTTor, TTE and WarranTTor are registered trademarks or trademarks of SafeTTy Systems Ltd in the UK and other countries.

What is the SafeTTy Certified training programme..

The **SafeTTy Certified** programme is designed to help people obtain the **skills** needed to develop safe, reliable and secure embedded systems using TT architectures – and provide **evidence** that they possess such skills.

The programme starts at “Level 0” and progresses to “Level 2”

In our view:



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- **Level 0** represents the foundation level qualification that should be expected from an individual desirous of learning the art of developing real-time embedded systems.
- **Level 1** represents the minimum qualification level that should be expected from an individual involved in the development of real-time embedded systems that are intended to be reliable, but where failure has no direct safety implications.
- **Level 2** represents the minimum qualification level that should be expected from an individual involved in the development of real-time embedded systems that may have some **safety implications** (for example, designs developed in compliance with IEC 61508 or in compliance with ISO 26262), and / or where system failure could prove **very expensive** (e.g. deep-sea products, satellite systems).

Various Courses offered under the SafeTTy Certified Track

Course Name: Level-0 Embedded C (EC)

Course Location: On-Campus

Course delivery: Face-to-face on-campus delivered by our internationally certified trainer.

Course Duration: 60hrs (4hrs. per day), conducted as per the convenience of the institution.



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Benefits for the course participants: Every participant will get a TI MSP432 launchpad kit along with Code Composer Studio. Course delivered on-campus. Internationally designed Courseware which is the copyright of SafeTTy Systems(UK) designed by Dr. Michael J. Pont. **Course focuses on the design aspects with Time Triggered architecture and not Event Triggered(Interrupt driven) architecture which is generally followed in engineering courses.** Internationally recognized Certificate will be awarded to successful candidates at the end of the course by SafeTTy Systems UK. Course is mapped to regular course curriculum using ARM Cortex M4 microcontroller. Further course levels are available for further study to craft a career in embedded systems domain. International Certificate for Level-0 from SafeTTy Systems.

Key features of this course

- A consistent focus on designing code that is safe and reliable.
- Course begins with the study of ARM architecture.
- A crash course on 'C' is included in the course itself, in case your skills are a little rusty.
- Creating a Simple Embedded OS; this is the starting point in learning about TT architectures.
- Learning about a single interrupt approach to carry out different tasks.
- Internationally-recognized certificate provided for successful candidates.

Course Objectives

After completing the course, participants should:

- Understand ARM architecture
- Use features of 'C' to create structured embedded code.
- Write code that is reliable and responsive.
- Be comfortable with designing code that adheres to timing constraints.
- Understand the necessity of an Embedded OS.
- Understand the 'One Interrupt per Microcontroller Rule' in the design of TT systems.
- Understand the basic building blocks of TT architecture.
- Be ready to go to the next certification level LEVEL 1.

Prerequisites



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Familiarity with Windows OS, good knowledge of binary number system and digital electronics. Familiarity with 'C' language would be an advantage.

Syllabus

- Overview of ARM architecture.
- Memory map of the ARM Cortex-Mx microcontroller.
- Timers of MSP432.
- Machine cycle and oscillator frequency calculations.
- SysTick Timer and its use in an Embedded OS
- Crash Course on C programming with focus on loops, functions, function prototypes, variable declaration and definition , pointers , typedefs , macros ,unsigned numbers .
- CCS environment.
- Creating project, building the target and running the code on target board.
- Understanding Energy trace in CCS.
- Reading Switches,Controlling Multi-colored LED
- Structuring Code and Object Oriented programming with C.
- Real Time constraints
- Creating a small Embedded OS.
- Multistate systems and function sequences.
- Generating PWM signal.
- Traffic signal system using FSM.
- Watchdog timer and their use in recovery from failure.
- Serial Port programming of MSP432.



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Mode of assessment for CERTIFICATION

- Closed book exam with university-style conditions.
- Planned exercises that would be submitted by candidates during the course.

Certification

- On successful completion of the course candidates will receive LEVEL 0 internationally recognized Certification from SafeTTy Systems.

Course duration and delivery

- Course duration is 60 hrs. (4hrs. per day).
- Course lectures will be held as per the convenience of the institution.
- Candidates are expected to complete 5 practical exercises .To be submitted during the course.
- Exercises carry weightage towards final certification.
- Final written examination of 2 hours' duration would be conducted on completion of the course.
- Successful candidates will receive SafeTTy Certified certificates within 2-months of successful course completion.

Course Name: Level-1 Foundations of Reliable Embedded Systems(TTa)

Course Location: On-Campus



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Course delivery: Face-to-face on-campus delivered by our internationally certified trainer.

Course Duration: 8-weeks, as per the convenience of the institution.

Benefits for the course participants: Course delivered on-campus. Internationally designed Courseware which is the copyright of SafeTTy Systems(UK) designed by Dr. Michael J. Pont. **Course focuses on the design aspects with Time Triggered architecture and not Event Triggered(Interrupt driven) architecture which is generally followed in engineering courses.** Internationally recognized Certificate will be

awarded to successful candidates at the end of the course. Course focuses on CPU schedulers. Further course levels are available for further study to craft a career in embedded systems domain. International Certificate for Level-1 from SafeTTy Systems.

Key features of this course

- A consistent focus on the creation of reliable embedded systems in a wide range of sectors.
- Provides an introduction to modern Time-Triggered (TT) system architectures.
- Provides a solid foundation for people interested in the development of safety-related systems.

Course Objectives

- After completing the course, participants should:
- Understand the key concepts of “real-time” systems and the importance of timing behavior when creating reliable embedded systems.
- Understand key design and programming techniques which can help to improve the reliability of embedded systems for use in a range of different market sectors.
- Understand time-triggered (TT) software architectures.
- Understand how to test and debug existing embedded systems.
- Be ready for more advanced training in software design and system verification and progress to LEVEL 2.

Prerequisites

- Level 0 Certification or Experienced Professionals wishing to learn about TT architectures.

Syllabus

- What is an Embedded and a Deeply Embedded System.
- Discussion on 8051 architecture.
- Why use C?



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- Difference between writing code for desktop and an embedded system and the differences in their software architecture.
- What are TASKS?
- What is a Real Time Embedded System?
- Accuracy of Loop Delays and Hardware Delays.
- What are Sandwich delays and why do we use them?
- What are timeout mechanisms and why and when do we need to timeout?
- What is the concept of balanced code?
- Concept of Worst Case Execution Time (WCET) and Best Case Execution Time (BCET) of Tasks.
- Understanding Interrupts and Interrupt response time.
- Understanding TT embedded operating system.
- Explore Co-operative and Hybrid schedulers and why we need them and when to employ them.
- How to handle Task Jitter?
- Timing analysis of tasks
- Compare 8-bit 8051, 32-bit ARM, soft processor implemented on FPGA and x86 platforms and how the performance is affected in each case. Which platform is reliable if employed in aerospace systems?
- What are multitasking systems?
- Running multiple tasks on a single CPU system with Task preemption. Here we learn about implementing a system employing a Hybrid scheduler. We also explore why a commercial RTOS may not be a good choice in systems where there are tight timing constraints.
- What is Priority Inversion and how to avoid it using Hybrid schedulers?
- What are distributed systems and their interconnection topologies?
- How are tasks running on different nodes synchronized in distributed systems?
- CAN protocol and its use in distributed systems.

Mode of assessment for CERTIFICATION



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- Closed book exam with university-style conditions.
- 12 Planned Exercises that would be submitted by candidates during the course. The exercises are critical in refining the skills learnt during the course.

Certification

- On successful completion of the course candidates will receive LEVEL 1 internationally recognized Certification from [SafeTTy Systems](#).

Course Duration and Delivery

- Course duration is 8 weeks.
- Course lectures will be held once a week for 4 hours with 2 hours of theory and 2 hours of practical's.
- Candidates are expected to complete the 12 exercises by the 12th week, before the written exam.
- Final written examination of 2 hours duration would be held at the end of the 15th week since the start of the course.
- Successful candidates will receive SafeTTy Certified certificates within 5 months from the start of the course.

Course Name: Level-2 Engineering of Reliable Embedded Systems(TTb)



Course Location: On-Campus

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Course delivery: Face-to-face on-campus delivered by Dr. Michael J. Pont.

Course Duration: 5 days (1 seminar every day for 3 hours followed by practical session for the rest of the day). Final written Exam will be conducted 12-weeks after the course completion.

Hardware board to be used: As per the institution's requirement. Board to be provided to the candidates by the institution. The details of the hardware board to be used should be shared with us 2-months prior to the course

commencement date so as to customise the software for the selected board.

Batch Size: Ideally 20 candidates per batch with a maximum of 50 candidates per batch. Batch size to be confirmed a week in advance before the commencement of the course.

Who will be the Course participants: Level-1 certified candidates.

Other candidates need to provide CV and relevant industry experience details 15-days in advance so that we can assess their suitability for this course in order that they can complete the course comfortably considering the intensiveness of the course.

Certification: 70% pass score in the 2-hour university style written examination conducted at the end of 12-weeks. Successful candidates receive Level-2 SafeTTY Certified certification within 2-months of successful course completion.

Benefits for the course participants: Course delivered on-campus. Internationally designed Courseware which is the copyright of SafeTTY Systems(UK) designed by Dr. Michael J. Pont. **Course focuses on the design aspects with Time Triggered architecture and not Event Triggered(Interrupt driven) architecture which is generally followed in engineering courses.** Internationally recognized Certificate will be awarded to successful candidates at the end of the course. Course focuses on CPU schedulers and international safety standards in different domains. International Certificate for Level-2 from SafeTTY Systems.

Detailed syllabus is available below.



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Session 1: Foundations of TT embedded systems

The material in Session 1 provides a foundation for the remainder of the course. We start by considering the differences between time-triggered and event-triggered system architectures. We consider a simple implementation option for time-triggered co-operative (TTC) software architectures. We consider how to design appropriate tasks for use with a TTC architecture. We consider multi-state systems. We consider some simple initial design examples.

Introduction to Exercise 1.

Session 2: TTC, TTH and TTP designs / Shared-Clock schedulers

In Session 2, we consider TT designs that support task pre-emption, including both “hybrid” designs (TTH) and fully-pre-emptive designs (TTP). We then consider some of the challenges involved in creating distributed embedded systems: that is, TT systems involving multiple processors connected in some form of network using (for example) a bus or star topology.

Introduction to Exercise 2.

Session 3: Designing and modelling your TT system

In Session 3, we first consider the modelling of TT systems. We consider the differences between synchronous and asynchronous task sets. We consider how to determine both the length of the major cycle, and the duration of the system initialisation phase. We then turn our attention to schedulability analysis and scheduler configuration for TT designs. We demonstrate how to create a ‘Tick List’. We demonstrate how to use the Tick List to determine the response times, processor loading and task jitter.

Introduction to Exercise 3.

Session 4: Monitoring your TT system

In Session 4, we demonstrate the effectiveness of appropriately-matched run-time monitoring systems when used with TT designs. We consider techniques for monitoring task execution times (both ‘Worst Case Execution Time’ – WCET – and ‘Best Case Execution Time’ – BCET). We also consider the importance of task-sequence monitoring. We consider use of Task Contracts and System Contracts as design aids. We go on to consider multi-mode systems. We consider the challenge of ensuring safe transitions between modes. We discuss the use of (and design of) ‘same but different’ tasks for use in systems that require independence (including designs involving ‘ASIL decomposition’ and / or ‘backup tasks’).

Introduction to Exercise 4.

Session 5: Conventional real-time operating systems (and TT alternatives)

We begin Session 5 by exploring in more detail some of the challenges involved in designing and implementing systems in which more than one task must run simultaneously. We first review ways in which we can approximate multi-tasking systems using a single CPU and a conventional “real-time operating system” (RTOS). We consider rate-monotonic and deadline-monotonic scheduling algorithms. We consider the use of mutexes and semaphores. We demonstrate the impact that priority inversion (PI) has on temporal determinism in all ET systems. We consider some partial solutions to PI problems through the use of “ceiling protocols”. We revisit TTH and TTP designs and demonstrate how we can eliminate the problems of PI completely using TT architectures.

Session 6: Recommended TT Platforms, quasi-TT designs, and a detailed case study

In Session 6, we begin by presenting a set of recommended ‘TT Platforms’ for use in reliable embedded systems (including those that must be developed in compliance with international safety standards, such as IEC 61508, ISO 26262 and DO-178). We explore ways in which we can employ a quasi-TT approach in designs that need to support additional (asynchronous) interrupts. We then demonstrate how to apply TT design approaches by tackling a significant (and realistic) case study.



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Skill Certification for Outcome-based Professional Education

[Home](#) > Management Module from SPJIMR for SPIT Students

Management Module from SPJIMR for SPIT Students

[Registration Open](#)

***This module is for students of Sardar Patel Institute of Technology only.**

Bhavans, Sardar Patel Institute of Technology (SPIT) has been granted academic autonomy from the academic year 2017-18. The Institute offers four undergraduate programs B.Tech Electronics, Electronics and Telecommunication, IT and Computer Engineering. The Institute also offers M.Tech and Ph.D programs.

SPIT is revamping the curriculum and organizing additional elective courses to enhance managerial and functional skills in emerging areas.

As part of this and as requested by SPIT, SPJIMR will be offering a 6 module Management Program. The number of participants for this stream will be 60.

The participants will be offered 1 module per semester. The program details will be as under.

Courses –

1. Economics and Business Environment
2. Marketing Management
3. Financial Management and Management Accounting



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4. Operations – Supply Chain Management
5. IT for Business
6. HR and Organisational Behaviour

Course design –

1) Each course will have 40 hours of content for the participants

1. Contact sessions – 20 hours (18 sessions of 70 minutes each)
2. Course pre-read – 10 hours
3. Assignment/group work/project etc – 10 hours

SPJIMR will design the content, deliver the program and will do the evaluation and certification.

2) Evaluation – will have in general 3 components

- | | |
|--|-----|
| 1. Attendance, participation, Class assignment | 20% |
| 2. Quiz/Mid-terms/ End terms/ Examinations | 60% |
| 3. Group work / Project Work | 20% |

3) Sessions will be scheduled as under

- A) 4pm to 6pm from Monday to Friday

or

- B) 10am to 1pm on Saturdays

Total 6 sessions (7 hours) per week, so that 1 course can be completed in 1 month.



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NPTEL/SWAYAM

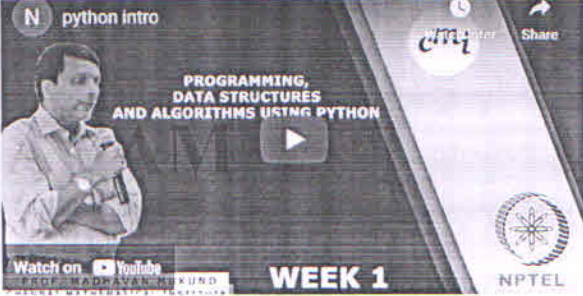
Programming, Data Structures And Algorithms Using Python	noc19-cs40	SWAYAM(Online)
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[Initiatives](#) [Programs](#) [NPTEL STARS](#) [NPTEL Testimonials](#) [More](#) [Log in](#)

Course Details

- Week 1: Introduction
- Week 2: Basics of Python
- Week 3: Lists, inductive function definitions, sorting
- Week 4: Sorting, Tuples, Dictionaries, Passing Functions, List Comprehension
- Week 5: Exception handling, input/output, file handling, string processing
- Week 6: Backtracking, scope, data structures, stacks, queues and heaps

N python intro



Share

Watch on YouTube

WEEK 1

NPTEL

Course Abstract

This course is an introduction to programming and problem solving in Python. It does not assume any prior knowledge of programming. Using some motivating examples, the course quickly builds up basic concepts such as conditionals, loops, functions, lists, strings and tuples. It goes on to cover searching and sorting algorithms, dynamic programming and backtracking, as well as topics such as exception handling and using files. As far as data structures are concerned, the course covers Python dictionaries as well as classes and objects for defining user defined datatypes such as linked lists and binary search trees.

Course Duration
Jan-Mar 2024

Enrollment
2023-11-09 to 2024-01-29

Exam Registration
to 2024-02-16

Exam Date
2024-03-24

Python for Data Science - Online	noc19-cs59	SWAYAM(Online)
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Course Details

- Week 1
- Week 2
- Week 3
- Week 4
- Supporting material for Week 4

About Course

Downloads

TA List

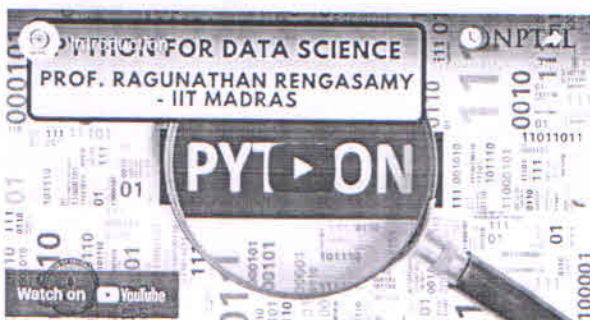
Statistics

Toppers List

Certificate Type

Feedback

Syllabus



Course Duration
Jan-Feb 2024

Enrollment
2023-11-09 to 2024-01-29

Exam Registration
to 2024-02-16

Exam Date
2024-03-24

Course Abstract

The course aims at equipping participants to be able to use python programming for solving data science problems.

Indian Economy: Some Contemporary Perspectives | noc21-hs51 | SWAYAM(Online)

Course Details

- WEEK 1
- WEEK 2
- WEEK 3
- WEEK 4
- WEEK 5
- WEEK 6
- WEEK 7
- WEEK 8
- Live Session



Course Duration
Jan-Mar 2024

Enrollment
2023-11-09 to 2024-01-29

Exam Registration
to 2024-02-16

Exam Date
2024-03-23

Course Abstract

Understanding the role of economic activities has always been a challenging task for non-economics students and it is an unavoidable task. This course is designed for undergraduate students to learn the art of understanding the functioning of the economy in the light of its mixed economic set up, historical developments and recent reforms. The course touches upon various relevant topics which are indeed useful for the budding economist and non-economics students who want to apply for general competitive exams and masters program in management and civil services.

Speaking Effectively | noc21-hs05 | SWAYAM(Online)

Schaudhary



Course Details

- Week 1
- Week 2
- Week 3
- Week 4
- Week 5
- Week 6
- Week 7
- Week 8
- Live Session

Course Duration:
Jan-Mar 2024Enrollment
2023-11-09 to 2024-01-29Exam Registration
to 2024-02-16Exam Date:
2024-03-23

Course Abstract

This course aims to introduce learners to the dynamics of effective spoken communication by establishing speaking as an autonomous medium with a distinctive vocabulary, syntax, structure, style and register. It will enable learners to participate in one to one interactions, in small groups and before a group. Learners are expected to master the fundamentals of speaking such as vocabulary, body language, pronunciation and basic conversation skills before they move on to more advanced activities such as appearing in interviews, making formal presentations and participating in meetings.

Coursera

Introduction to Git and GitHub (18 hours) | Coursera

G Introduction to Git and GitHub

Enroll for Free
Starts Feb 20

About

Outcomes

Modules

Recommendations

Testimonials

Reviews

Read more

Introduction to Version Control

Module 1 • 9 hours to complete

Using Git Locally

Module 2 • 6 hours to complete

Working with Remotes

Module 3 • 7 hours to complete

Collaboration

Module 4 • 6 hours to complete

Instructor

Instructor ratings 4.9 ★ (2,001 ratings)



Top Instructor

Google Career Certificates

Google

323 Courses • 8,746,120 learners

Offered by



Google

Learn more



Schaudhary



Getting and cleaning data (21 Hours) Coursera

For Individuals For Businesses For Universities For Governments

coursera

Explore

What do you want to learn?

Online Degrees

Find your New Career

English

Browse > Data Science > Data Analysis



Getting and Cleaning Data

This course is part of multiple programs. [Learn more](#)

Taught in English | [20 languages available](#) | Some content may not be translated



Instructors: [Jeff Leek, PhD](#) +2 more

Enroll for Free
Starts Feb 14

Financial aid available

Course

Gain insight into a topic and learn the basics.

4.5 ★ (8,044 reviews) | 12

19 hours (approximately)

Flexible schedule

Learn at your own pace

[View course modules](#)

There are 4 modules in this course

Before you can work with data you have to get some. This course will cover the basic ways that data can be obtained. The course will cover obtaining data from the web, from APIs, from databases and from colleagues in various formats. It will also cover the basics of data

[Read more](#)

Week 1

Module 1 • 2 hours to complete

Week 2

Module 2 • 1 hour to complete

Instructors

Instructor ratings 4.3 ★ (325 ratings)



[Jeff Leek, PhD](#)

Johns Hopkins University

32 Courses • 1,607,983 learners

[View all 3 instructors](#)

Google Crash Course on Python - 23 Google(Coursera)



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coursera

Explore

What do you want to learn?

Online Degrees

Find your New Career

English



Browse > Information Technology > Support and Operations

Google

Crash Course on Python

This course is part of [Google IT Automation with Python Professional Certificate](#)

⌕ Taught in English | [Video subtitles available](#)

G Instructor: [Google Career Certificates](#)
Top instructor

Enroll for Free
Starts Feb 20

Financial aid available

Course

Gain insight into a topic and learn the fundamentals.

4.8 ★ (35,252 reviews) |

Beginner level

No previous experience necessary.

39 hours (approximately)

Flexible schedule

Learn at your own pace

[View course modules](#)



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Marketing Management							
		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1

Marketing has evolved as an applied discipline and its importance to any business venture cannot be underestimated. This course refers to how marketing creates value for a firm's chosen customers and extracts a share of that value for the firm.

Marketing theory and practice are justified in the belief that customers use a product or service because they have a need, or because it provides a perceived benefit. The skill of marketing is the skill to monitor and understand customers, competitors, and collaborators, and to find in each domain a better way to design and deploy the firm's limited resources to serve its customers at a profit. In this way, marketing helps to set a firm's strategic direction.

Marketing, thus defined, is a broad general management responsibility and not something to be delegated to marketing specialists. Its skills are required in all careers concerned with the strategy of organizations. This course, therefore, deals with marketing in a variety of settings: in the old and new economy, service firms, industrial manufacturing companies, as well as manufacturers and retailers of consumer products.

It is very important to bear in mind that there are no clear-cut solutions to marketing-related problems. There are a variety of analytical approaches, tools, and techniques that we can use in thinking about strategic issues (that is, problems of such importance and urgency that they have organization-wide ramifications and can significantly undermine performance and viability). In the process, we will study these analytical approaches, apply them to cases, and make our own generalizations as to their general applicability and value with specific objectives.

Course Objective:

The objective of the course is to introduce the basics of marketing and develop an appreciation of its pervasive role in business, political and social life. The specific objective of the course is to help students:

1. Think strategically about marketing
2. Use key frameworks to analyze situations and make decisions
3. Understand how to create a strong marketing plan

Prescribed text: Marketing Management 15th edition by Philip Kotler and Kevin Lane Keller

Pedagogy:

The pedagogic mix will include concept discussions, cases, practical exercises, assignments, fieldwork and presentations. Emphasis will be on discussion and the active exchange of ideas. The course is intended to be a vehicle for self-learning and group learning. Carefully thinking out your position and displaying tenacity in defending it from attack by skeptics are both important requirements. A willingness to share your perspectives, knowledge, and experience is another important factor for a satisfactory classroom experience in this course.

The learning process will be inductive i.e. we will understand the concepts as applied to a real-life company. To ensure that learning occurs and doubts are clarified it is imperative that all designated pre-

Marketing Management							
		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1

class preparatory work is done and we are all prepared to spend time in class in clarification of doubts/issues. Ethical issues will be taken up where they are relevant.

Special Instructions:

1. This is not a course for spectators. It is a highly experiential/interactive course. **Thorough pre-class preparation is a MUST.**
2. All submissions are to be made before designated classes/ presentations.

Class Participation: Two main criteria will be used in reaching our judgment about your class participation:

1. Depth and Quality of Contribution

This will be the most important dimension of participation concerns what it is that you are saying. A high quality comment reveals depth of insight, rigorous use of case evidence, consistency of argument, and realism. To perform well on these criteria, it is important that you come to class with a definite action plan and be prepared to help your classmates appreciate the appropriateness of your plan/point.

2. Moving Your Peers' Understanding Forward

This dimension looks into:

Relevance — Effective listening, a good sense of timing, and a willingness to either integrate the work of others or challenge their ideas are the skills that lead to relevance.

Clarity and Conciseness — Clarity involves speaking with urgency, vividness, and persuasiveness. To be concise, make your best point in the shortest possible time.

Involvement — Involvement means thoughtful listening, concentration, tracking of the discussion, and a poised readiness to contribute selectively.

Suggestions for Reading beyond the Assigned Materials: A good general-purpose reference book is assigned as the textbook with relevant chapters marked in session details. Furthermore, your time is better spent thinking and applying concepts to current situations for which you can choose to read up newspapers or business magazines available in your library.

Assessment Plan: 3 Quizzes (30), Mid-term (20), Industry Project (50)

Marketing Management							
		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1

Industry Project

You need to analyze the marketing strategy for the assigned brand and address the following:

1. Highlight the consumer decision-making process the customers undertake to purchase the brand. (10 marks)
2. Identify the market segment(s) and the variable(s) for segmentation. (10 marks)
3. Identify the target market for the offering and give reasons why? (10 marks)
4. State and critically evaluate the positioning of the brand. Also, (30 marks)
 - a. write a positioning statement for the brand, and
 - b. draw a perceptual map
5. Recommend a new product line addition for the company's product portfolio. (10 marks)
6. Suggest the pricing for your brand. (10 marks)
7. Which channels will you use to market your brand? (10 marks)
8. What are the different ways in which you can promote your brand? (10 marks)

Methodology

- a) Conduct a primary research among 15 consumers/users to understand their purchase process and decision roles. Also, analyze their perceptions regarding the brand and the competition.
- b) Primary data: Minimum field time for interviewing & data collection = approx. 2.5 hours/ student.
- c) Secondary data: Printed material/ company's websites/ company personnel interview/ reports.

Other details

- a) The project scores will be weighted down to 50%.
- b) The submission should be in .pptx- format and total number of slides should not exceed 15.



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Marketing Management							
		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1

COURSE PLAN

Each module has a structure that will be followed right through the course. The specified text for the relevant sessions should be read before each class.

Module 1: MARKETING AS A VALUE CREATION PROCESS

Learning Objectives-

- Concept of needs, wants and demands
- Concept of value and satisfaction
- Ansoff's Grid
- 5 Cs
- SWOT Analysis
- Marketing as a department & Marketing as a function

Session	Themes	Activity in the class	Required Reading from the textbook
1	Topic: Introduction to Marketing	Discussion	Chapter 1
2	Topic: Marketing Strategy	Concept discussion	Chapter 2
Module 1- Key Concepts Covered Marketing Process, 5Cs and SWOT Analysis			

Module 2: UNDERSTANDING CONSUMERS

Learning Objectives

- The need for understanding consumer behavior
- Consumer Decision Process
- Business Buying Process



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Marketing Management							
		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1

Session	Themes	Activity in the class	Required Reading from the textbook
3	Topic: Customer Relationship Management	Concept discussion	Chapter 3
4	Topic: Understanding Consumer Behavior	Concept discussion	Chapter 6
5	Topic: Understanding Business Buying Behavior	Concept discussion	Chapter 7
6	Topic: Understanding Global Markets	Concept discussion	Chapter 8

Module 2- Key Concepts Covered

Understanding consumer needs and differences among customers. Appreciating the differences between consumer buying and industrial buying. Customer lifetime value and customer equity.

Module 3: CHOOSING YOUR CUSTOMERS

Learning Objectives

- Need for segmenting, targeting and positioning
- Bases for segmenting consumer and business markets
- Requirements for effective segmentation
- Developing a positioning strategy
- Points-of- difference and Points-of-Parity
- Brand equity
- Branding strategies – Branding decisions, extensions and portfolio

Marketing Management							
		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1

Session	Themes	Activity in the class	Required Reading from the textbook
7	Topic: Segmentation	Concept discussion	Chapter 9
8	Topic: Targeting	Concept discussion	Chapter 9
9	Topic: Brand Positioning	Concept discussion	Chapter 10
10 & 11	Topic: Brands and Brand Equity	Concept discussion	Chapter 11

Module 3 – Key Concepts Covered

Segmentation, Targeting, Positioning, Branding

Session 12: Mid-Term Presentation**Module 4: Marketing Research**Learning Objectives-

- Marketing Research Process
- Framing suitable questions
- Collecting representative Data
- Ways to analyze data

Session	Themes	Activity in the class	Required Reading from the textbook
13 & 14	Topic: Marketing Research	Concept discussion	Chapter 5

Module 4 – Key Concepts Covered

Marketing Research Question, Secondary and Primary Data, Questionnaire

Marketing Management							
		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1

Module 5: CREATING A VALUE PROPOSITION- DESIGNING YOUR OFFER

Learning Objectives-

- Product characteristics and classifications
- Product mixes
- Product line analysis and Product line-length
- Strategies for the PLC

15 & 16	Topic: Creating Value – Product Policy and PLC	Concept discussion	Chapters 13 & 14
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Module 5 - Key Concepts Covered

Creating value through product benefits, features, product line; Creating value over the PLC

Module 6: DELIVERING VALUE- Going to Market

Learning Objectives

- Functions of marketing intermediaries
- Significance of various types and levels of channels
- Channel design decisions
- Channel integration and systems
- Channel management including managing channel conflict

Session	Themes	Activity in the class	Required Reading from the textbook
17 & 18	Topic: Delivering Value – Going to market	Concept discussion	Chapters 21 & 22

Module 6 – Key Concepts Covered

Channel levels, Channel Design, Channel system and architecture, Channel conflict

Marketing Management							
		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1

Module 7- COMMUNICATING THE VALUE PROPOSITION

Learning Objectives

- 6M framework
- Communication vehicles
- Developing effective communication
- Deciding on communication mix

Session	Themes	Activity in the class	Required Reading from the textbook
19 & 20	Topic: Communicating Value	Concept discussion	Chapters 17, 18, 19, 20

Module 7- Key Concepts covered

6M Framework, Integrated and Marketing Communication tools

Module 8: CAPTURING VALUE

Learning Objectives

- Factors affecting pricing
- Basic principles of pricing, including value pricing
- Price sensitivity, Price customization

Session	Themes	Activity in the class	Required Reading from the textbook
21 & 22	Topic: Pricing to Capture value	Concept discussion	Chapter 16

Module 8 - Key Concepts Covered

Relationship between value, costs, and volume, price response estimation, price customization

23 & 24	Project Presentations		
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		Faculty	Dr. Rakhi Thakur, Dr. Sunny Arora	Sessions	24	Group work, Quizzes	1
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Schedule Plan for SPIT Mktg Mgmt course 2022

Session taken by	Date	Session	Time
Prof. Rakhi Thakur	5-Sep 2022	Session 1	5:40 PM - 6.50 PM
	6-Sep 2022	Session 2	5:40 PM - 6.50 PM
	7-Sep 2022	Session 3	5:40 PM - 6.50 PM
	12-Sep 2022	Session 4	5:40 PM - 6.50 PM
	13-Sep 2022	Session 5	5:40 PM - 6.50 PM
	14-Sep 2022	Session 6	5:40 PM - 6.50 PM
	19-Sep 2022	Session 7	5:40 PM - 6.50 PM
	20-Sep 2022	Session 8	5:40 PM - 6.50 PM
	21-Sep 2022	Session 9	5:40 PM - 6.50 PM
	26-Sep 2022	Session 10	5:40 PM - 6.50 PM
	27-Sep 2022	Session 11	5:40 PM - 6.50 PM
	28-Sep 2022	Session 12	5:40 PM - 6.50 PM
Prof. Sunny Arora	12-Oct 2022	Session 13 & 14	4:20 PM - 6.50 PM
	17-Oct 2022	Session 15 & 16	4:20 PM - 6.50 PM
	20-Oct 2022	Session 17 & 18	4:20 PM - 6.50 PM
	2-Nov 2022	Session 19 & 20	4:20 PM - 6.50 PM
	7-Nov 2022	Session 21 & 22	4:20 PM - 6.50 PM
	17-Nov 2022	Session 23 & 24	4:20 PM - 6.50 PM



S. Chaudhary

Fundamentals of Accounting & Finance

Audience: Final year engineering students

Credit: 1.5

Sessions: 20

The accounting and finance function plays a major role in the economy and business of a country. There is a myth that the information related to finance is of use to only those people in organization who are in finance function. The truth is that more the understanding of finance information in various functions of the organization, more is the probability that the decisions are going to be more informed and decision making shall be faster.

Learning Goal:

This course aims at explaining the basic concepts of Accounting and Finance to participant of Engineering course so as to enable them to apply these concepts for more informed decision making in their future work environment. By the end of this course, the participants will be able to understand the broad aspects of accounting, costing, and Indian financial system.

Learning Objectives:

The participants shall learn the following in the course:

- a) Basics of Financial Accounting;
- b) Costs in an Organization and their use in decision making; and
- c) An overview of the financial system and markets

Book:

Finance Sense by Prasanna Chandra, 5th Edition, McGraw Hill Higher Education.

Handouts shall be provided before the class.

Session - wise Plan

Sr. No	Topics - contents	Remarks
1-2	Introduction to Financial Accounting	Chapter 1
3-4	Understanding different Types of Accounts: Assets, Liabilities, Income, Expense and Capital	Chapter 3
5-6	Introduction to Income Statement and Balance Sheet	Chapter 3
7-8	Making Sense of Income Statement and Balance Sheet Numbers – Ratio Analysis	Chapter 2
9	Introduction to Costs in Organizations	Chapter 5
10	Understanding Classification of Costs	Chapter 5
11-12	Understanding methods of Cost Absorption	Chapter 5
13-14	Using Cost for Decision Making- CVP Analysis	Chapter 6



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15-16	Using Costs for Decision Making- Average Costing and Marginal Costing	Chapter 6
17-18	Banks and non-banking financial companies	Chapter 9
19	Insurance Companies, Asset management Companies	Chapter 9
20	Financial Markets	Chapter 9

Evaluation:

Quizzes (2): 30 marks
Assignment: 40 marks
End term: 40 marks



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UX Integrated Program for B.Tech (all branches)

powered by ImaginXP

ImaginXP's UX Integrated Program for B.Tech has been the most sought after and an in-demand career oriented program, running successfully across many famous Universities in the country. ImaginXP has carved its niche in Digital Design and Future Skills Space and the industry centric curriculum team has handcrafted this program for engineering professionals who like to create a value and edge over other professionals in this challenging career times.

LEARNING OUTCOMES OF UX INTEGRATED COURSE

- a. Create people with expertise who will eventually emerge as leaders to influence future of UX design
- b. Produce professionals with good analytical, thinking and innovative abilities, skilled with specific techniques, creative in their solutions and knowledgeable about latest development in the field of user experience
- c. The student after undergoing this course will be able to do user research, persona mapping, customer journey mapping, creating Wireframing and prototyping, user interface designing, empathy map, user research methodologies, questionnaire building, task flow, as is and to be task flow, card sorting, building information architecture, visual design for different form factors, usability testing techniques.
- d. The student will learn design process, way of thinking, strategic thinking in design field and design thinking
- e. Acquire practical expertise in the field of problem solving leading to innovation
- f. Become proficient in empathizing with users and analysing their needs
- g. Enable Students to go for higher studies in the area of their specialization.

CONDUCT PATTERN OF THE UX SPECIALIZATION

IMAGINXP teaches 4 subjects of UX Design and conducts a project. This is skill based training and therefore, focus will be on practical learning.

S. No.	Name of Subject	Credit	Hours
1	UX Design & Digitalization	4	60
2	Empathy & Its Tools	3	45
3	User Research & Its Application	4	60
4	Design Thinking & Its Applications	4	60

RESPONSIBILITIES

A) For IMAGINXP

- a. Shall ensure quality delivery of the courses by qualified and trained faculty in online mode
- b. Will be responsible for setting question papers, evaluation of exam paper and evaluation of project work.
- c. Assessments & continuous evaluation will be conducted by IMAGINXP faculty
- d. Shall provide internship support to best 5-7% of meritorious performers on the basis of evaluation and design score
- e. Shall be responsible for preparing all reading and training material for all subjects.

B) For the University

- a. Registration of students for the batch.
- b. Fee collection and payment to IMAGINXP



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- c. University will be responsible for local logistics, lodging and boarding of the IMAGINXP trainers and other staff in case of campus sessions
- d. University will ensure availability of said hours for the conduct of UX Integrated course
- e. Basic Infrastructure —
Classrooms with Audio Video Facilities and White Board
Internet Facility
Computer Lab with Computer systems with minimum i5 processor
Printing of worksheets
Other required stationary material
Online learning platform and tools in case of online classes

BATCH SIZE & STUDENT INTAKE

Minimum 30 students.

CERTIFICATION

Certificate to eligible students will be awarded by IMAGINXP

FEES

Course of 4 papers spread in 3 years (from semester IV till semester VII - 1 paper each sem).
The cost of that will be –

Rs. 25,000/- for complete course (keeping SPIT BTech fees structure in mind)

SYLLABUS

Subject 1: UX Design & Digitalization

CODE: UXDD | CREDITS: 4

1. Understand the evolution of UX design as an industry practice and learning about UX industry experts.
2. Understanding UX design processes and methodologies—user centred design, 5S model. Job roles and responsibilities in the UX industry. 6D UX Process
3. UX industry trends. Deep-dive in UX methodologies, case studies in UX design, heuristic evaluation, understanding product UX lifecycle.
4. Project on UX design implementation with industry relevant problem statement. Understand how technology and digitalization is transforming different industry segments – BFSI, manufacturing, retail, automotive, media, FMCG, logistics, oil & gas. Learn how to understand industry specific problems and user needs and design experiences for different industries.
5. Research and design for all industry segments using a toolkit. Digitalization for the bottom of the pyramid. Localization of experience.
6. Wireframing & Prototyping - Practice based module to learn the tools required to design wireframes and prototypes. Design wireframes on paper and translate paper concepts into digital wireframes and Understand and practice the techniques involved in designing digital wireframes for web, mobile, wearables, HMI and other digital screens. Understand and practice the techniques involved in creating digital prototypes.

Subject 2: Empathy & Its Tools

CODE: ET | CREDITS: 3

1. Learn how to understand users, techniques to empathize with users and identify key user problems.
2. Learn how to gain insights from empathy and define problems statements.
3. Empathy tools – techniques for getting empathy insights through interviews, empathy maps,



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emotional mapping, observation, Understand the users, user's interaction with the environment, people and culture, UX and societies, creating ethnography mood boards, user scenarios, storyboard, ethnography and user research, understanding research problems, data gathering techniques

4. Perform field study to understand people design
5. Project submissions empathy mapping

Subject 3: User Research & Its Application

CODE: UR | CREDITS: 4

1. User research importance, KPIs in User Experience, user research goals, heuristics analysis, user segmentation, user personas, identifying and recruiting users for the research, preparing a questionnaire for user research, user research methodologies – Qualitative and Quantitative analysis, user interviews, focused group discussion, expert reviews, tools for user research.
2. Understanding cognitive psychology and user behaviour. Application of user research, building a user lab, creating a user day, constant user feedback, listening posts and user research, understanding user problems, empathy map
3. Performing user research in the ecommerce industry. Performing user research in the insurance industry. Performing a user research with 20 users on a chosen problem

Subject 4: Design Thinking & Its Applications

CODE: DTA | CREDITS: 4

1. Introduction to design thinking, history of design thinking, wicked problems, case studies in design thinking, design thinking process, implementing the process in driving innovation, design thinking in social innovations
2. Tools of design thinking – persona, customer journey map, AS-IS, TO-BE Processes, product lockdown workshops
3. An exercise in design thinking – implementing design thinking for making the process of a user better. Student to choose one industry segment to implement design thinking process. Design thinking case studies in retail, design thinking case studies in banking, design thinking case studies in management decisions
4. Design thinking process and implementing it for a digital product



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Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Autonomous Institute Affiliated to University of Mumbai)

Department of Computer Engineering

CURRICULUM SCHEME FOR MINOR ACADEMIC PROGRAM AT SPIT

2020 ITERATION: **Computer Engineering**

Sem IV								
No	Type	Course	L	T	P	O	E	C
1	S/M	Data Structures and Algorithms	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem V								
No	Type	Course	L	T	P	O	E	C
1	S/M	Database Management Systems	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem VI								
No	Type	Course	L	T	P	O	E	C
1	S/M	Operating Systems	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem VII								
No	Type	Course	L	T	P	O	E	C
1	S/M	Web Technology/ Machine Learning	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Registration Fees for Minor in Computer Engineering is Rs. 24,000/- (Rs. 6000 per course) for SPIT students

Registration Fees for Minor in Computer Engineering is Rs. 32,000/- (Rs. 8000 per course) for Non SPIT students



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Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous Institute Affiliated to University of Mumbai)
Department of Computer Engineering

Data Structures and Algorithms–

Course Contents:

1. Random-access-machine model, asymptotic behavior of time/space complexity.
2. Elementary data-structures: arrays, lists, queues, stacks and their applications.
3. Binary search algorithm, binary trees, binary-search-tree data-structure.
4. Balanced binary-search-tree: Red-Black trees.
5. Hashing data structure.
6. Heap data structure.
7. disjoint-set union structure
8. Sorting algorithms, Greedy paradigm, Divide and conquer, Dynamic-programming paradigm
9. Data structures for graphs: adjacency lists, adjacency matrix.
10. Graph algorithms: Depth First Search, Breadth First Search, Minimum Spanning Tree
11. Single-source shortest path computation, topological sorting of a partially ordered set, convex-hull computation.

Database Management Systems –

Course Contents:

1. Relational model, Relational query languages, Relational algebra, Tuple and domain calculus
2. Structured Query Language: structure, Join expressions, views.
3. E-R model, Database integrity, Triggers.
4. Functional Dependency theory, Normal forms, algorithms for decomposition
5. Row-wise and column database, database buffering. Indexing, B+-tree indices, hashed files, bitmap indices, R-trees.
6. Query Processing
7. Query Optimization
8. Transactions, ACID properties, Concurrency Control, Recovery
9. Parallel Databases: I/O parallelism, Inter-query and intra operation parallelism.
10. Distributed Databases: Storage, distributed transactions, commit protocols, concurrency control in distributed databases.

Operating Systems –

Course Contents:

1. Introduction: Goals & Structures of O.S, Basic functions, Interaction of O.S. & hardware architecture, Batch, Multiprogramming. Multitasking.
2. Process and Threads Management: Process Concept, Process states, Process control, Threads, Types of scheduling, Scheduling Algorithms, System calls.
3. Concurrency: Principles of Concurrency, Mutual Exclusion: S/W approaches, H/W Support, Semaphores, Pipes, Message Passing, Signals, Monitors.
4. Inter Process Communication: Race Conditions, Critical Section, Mutual Exclusion, Hardware Solution, Strict Alternation, Peterson's Solution, The Producer Consumer Problem, Semaphores, Event Counters, Monitors, Message Passing, Classical IPC Problems: Reader's & Writer Problem, Dining Philosopher Problem etc., Scheduling, Scheduling Algorithms.
5. Deadlock: Principles of Deadlock, Starvation, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, System calls.



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Department of Computer Engineering

6. Memory Management: Memory Management Requirements, Fixed and Variable Memory Partitioning, Memory Allocation Strategies, Paging and Fragmentation. Demand Paging, Security Issues. Virtual Memory : Concepts, VM management, Page Replacement, Thrashing.
7. I/O Management & Disk scheduling: I/O Devices, Organization of I/O functions, Operating System Design issues, I/O Buffering, Disk Scheduling, RAID
8. Unix/Linux Operating System: Development Of Unix/Linux, Role & Function Of Kernel, System Calls, Elementary Linux command & Shell Programming, Directory Structure.

Web Technology –

Course Contents:

1. Introduction: Concept of WWW, Internet and WWW, HTTP Protocol : Request and Response, Web browser and Web servers, Features of Web 2.0
2. Web Design: Concepts of effective web design, Web design issues including Browser, Bandwidth and Cache, Display resolution, Look and Feel of the Website, Page Layout and linking, User centric design, Sitemap, Planning and publishing website, Designing effective navigation
3. HTML: Basics of HTML, formatting and fonts, commenting code, color, hyperlink, lists, tables, images, forms, XHTML, Meta tags, Character entities, frames and frame sets, Browser architecture and Web site structure. Overview and features of HTML5
4. Style sheets: Need for CSS, introduction to CSS, basic syntax and structure, using CSS, background images, colors and properties, manipulating texts, using fonts, borders and boxes, margins, padding lists, positioning using CSS, CSS2, Overview and features of CSS3
5. JavaScript: Client side scripting with JavaScript, variables, functions, conditions, loops and repetition, Pop up boxes, Advance JavaScript: Javascript and objects, JavaScript own objects, the DOM and web browser environments, Manipulation using DOM, forms and validations, DHTML : Combining HTML, CSS and Javascript, Events and buttons
6. XML: Introduction to XML, uses of XML, simple XML, XML key components, DTD and Schemas, Using XML with application. Transforming XML using XSL and XSLT
7. PHP: Introduction and basic syntax of PHP, decision and looping with examples, PHP and HTML, Arrays, Functions, Browser control and detection, string, Form processing, Files, Advance Features: Cookies and Sessions, Object Oriented Programming with PHP
8. PHP and MySQL: Basic commands with PHP examples, Connection to server, creating database, selecting a database, listing database, listing table names, creating a table, inserting data, altering tables, queries, deleting database, deleting data and tables, PHP myadmin and database bugs

Machine Learning –

Course Contents:

1. Preliminaries: Multivariate calculus, Linear algebra, Probability theory
2. Supervised and Unsupervised Learning
3. Model and feature selection, over-fitting and generalization, bias-variance tradeoffs
4. Optimization for machine learning: (stochastic/mini-batch) gradient descent
5. Deep learning: CNN, RNN, LSTM, auto-encoders
6. Structured output prediction: multi-label classification, sequence tagging, ranking
7. Ensemble methods: boosting, bagging, random forests
8. Recommendation systems: ranking methods, collaborative filtering via matrix completion



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Department of Computer Engineering

9. Kernel extensions for PCA, clustering, spectral clustering, manifold learning
10. Probability density estimation and anomaly detection
11. Time-series analysis and modeling sequence data
12. Sparse modeling and estimation
13. Online learning algorithms: Perceptron, Widrow-Hoff, explore-exploit
14. Statistical learning theory: PAC learning,



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Subject	Sessions	Hours/Week	Hours/ semester (14 weeks)	Credit
Wireframing & Prototyping	Lecture	2	42 hours/sem	3
	Practical			
	Tutorial			
	Studio Work	1		

Course objective:

- Demonstrate skills for low-fidelity & high fidelity wireframes and how design can be translated from concept to interactive prototype.

Knowledge

- Understand the difference between low fidelity & high fidelity wireframes

Skill

- Develop & demonstrate skills in Adobe XD
- Develop an understanding of wireframes and its significance

Synthesis

- Crafting seamless communication through enhanced user interfaces.

Course Content:

S No.	Hours	Content	Detail	Lecture	Practical
2	20	Introduction to Adobe XD	• Workspace & Tools, Drawing and editing graphics, cc libraries, Symbols • Images & vectors: How to import? Integration with Photoshop/Illustrator, Masking, Color, Creating Icons • How to make a gradient blur • Working with visual data/images • adding text fields and buttons • Creating 12 column card dashboard using repeat grid tool • How to make mobile phone mockup with software • Advanced prototyping using buttons and drop down menus • How to export images and code from a software for developers	8	12
1	10	Wireframe Basics	• Early design Iterations • Organising & Labelling the	2	8



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			content • Searching and Navigation • Layout • Develop Low fidelity wireframes - Paper prototype • How to draw wireframes on paper • Wireframe elements/styles/ • templates • Drawing the wireframe • Sitemap & Flow map		
3	12	Hi fidelity Wireframe	• Grids • Repeat Grids/advanced repeat Grid • Overlay, Responsive resize, Artboard size • Use web safe fonts • Creating custom Icons • How to create a mobile phone mockup with software	4	8

INDICATIVE ASSIGNMENT BRIEF AND EVALUATION MATRIX

End – Term Evaluation (Exam/Jury): 100 Marks	Brief : Students can choose any domain according to their interest create a high fidelity wire frame Create the complete prototype using Adobe XD.
EVALUATION PARAMETER / CRITERIA	
	Marks :100
	End Term Evaluation (Exam/Jury)

REFERENCE MATERIAL

Books

- Allanwood, G., Beare, P., 2014. Basics Interactive Design: User Experience Design. China: Bloomsbury Publishing
- Wood, D., 2014. Basics Interactive Design: Interface Design. China: Bloomsbury Publishing
- Grant, W., 2018. 101 UX Principles. Packt Publishing Limited

REFERENCE MATERIAL FOR STUDENTS



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Books

- Norman, D. The Design of Everyday Things: Basic Books Publishing
- Rogers, Y., Sharp, H., Preece, J. Interaction Design: Beyond Human-Computer Interaction: Wiley Publishing
- Tufte, E.R. Envisioning Information: Graphics Press
- Lazar, J., Feng, J.H., Hochheiser, H. Research Methods in Human-Computer Interaction: Wiley Publishing

INFRASTRUCTURE REQUIREMENT

- Well-equipped lab with Software Application installed
- Availability of relevant books in the Recourse Centre
- Faculty Laptop that supports Adobe XD software (Windows 10 & above/MacBook)



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Department of Electronics, Electronics & Telecom. Engineering

CURRICULUM SCHEME FOR MINOR ACADEMIC PROGRAM AT SP-IT 2020-V.1

Industrial Internet of Things Engineering

Sem IV								
No	Type	Course	L	T	P	O	E	C
1	S/M	Minor-I	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem V								
No	Type	Course	L	T	P	O	E	C
1	S/M	Minor-II	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem VI								
No	Type	Course	L	T	P	O	E	C
1	S/M	Minor-III	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem VII								
No	Type	Course	L	T	P	O	E	C
1	S/M	Minor-IV	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3



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Department of Electronics, Electronics & Telecom. Engineering

CURRICULUM SCHEME FOR MINOR ACADEMIC PROGRAM AT SP-IT

Industrial Internet of Things Engineering (I²oTE)

The Industrial Internet of Things Engineering minor introduces the concepts, tools and techniques that are involved in designing and programming electronic devices that involve sensing, computation, and communication. The minor prepares students to understand the design tradeoffs present in aggregating and processing information generated by these networked devices either locally or in the cloud. Students completing this minor will be ready to effectively compete in industries related to embedded and networked systems and IIoT technologies.

Sem IV								
No	Type	Course	L	T	P	O	E	C
1	S/M	Application Specific System Design	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem V								
No	Type	Course	L	T	P	O	E	C
1	S/M	Embedded "C" Programming & Real-time Software Development	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem VI								
No	Type	Course	L	T	P	O	E	C
1	S/M	Software Design for Discrete time Control Algorithms	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

Sem VII								
No	Type	Course	L	T	P	O	E	C
1	S/M	Industrial Internet of Things (IIoT) System design and Applications	2	0	2	4	08	3
		TOTAL	2	0	2	4	08	3

the department will accept introductory courses from other departments/ institutes as the prerequisites or If not done before starting of the course, then students needs to put additional 4 to 8 hours of work for completion of prerequisites.

the rules of registration, fees, evaluation & certification are as per SP-IT minor academic program rule book.



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(Autonomous Institute Affiliated to University of Mumbai)

Department of Electronics, Electronics & Telecom. Engineering

Minor-I: Application Specific System Design –

Prerequisites: Generic processor/controller architectures

Course Outcomes: After successful completion of the course, student will be able to

CO1: Understand the general process of embedded system development

CO2: Effectively interface the peripherals to microcontrollers

CO3: Know the Hardware-Software design issues and testing methodology

CO4: Design embedded systems as solutions to real-world problems

Course Contents:

Architectures of Embedded Systems, Programming models for Single-Core and Multi-Core architectures, Introduction to Hardware/Software Co-design. Embedded applications programs development environment, Real-time spectrum, Technology comparison, Popular Cores, Computational devices selection criteria, Real-time software development environment, Sensors & Transducers, Peripherals & their significance, Characteristics and Quality attributes of embedded systems, Minimum system design.

Minor-2: Embedded “C” Programming & Real-time Software Development –

Prerequisites: (a) Foundation of programming in C, (b) Basics of Memory hierarchy
(c) Fundamentals of Operating system

Course Outcomes: After successful completion of the course, student will be able to

CO1: Develop and maintain applications written using Embedded C.

CO2: Develop to configure device drivers

CO3: Evaluate the requirements of programming Embedded Systems, related software architectures and tool chain for Embedded Systems.

CO4: Understand the Real time operating system concepts and implement the RTOS development tools in building real time embedded systems

Course Contents:

Comparison of Embedded “C” with ANSI C, Programming model (small, medium, large, huge), Memory types (Flash, EEPROM, RAM) Multiple memory areas/banks (partition), Hardware initialization (I/O register mapping). Standard I/O and Preprocessor Functions.



Signature



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Autonomous Institute Affiliated to University of Mumbai)

Department of Electronics, Electronics & Telecom. Engineering

Significance of main () function, variables definition, data type declaration, conditional statements (if, switch, case), loops (while, for), functions, arrays and strings, structures and union, bit operations, macros, unions, etc.

Real-time generic architecture, Process and Scheduling Systems with various algorithms, Free RTOS Task Management and Real Time Scheduling, Free RTOS- Inter Task Communication.

Minor-III: Software Design for Discrete time Control Algorithms –

Prerequisites: (a) Control System Basics, (b) Numerical Methods

Course Outcomes: After successful completion of the course, students will be able to

CO1: Design a PI / PID Controller for a specific Industrial Applications

CO2: Write Discrete time equations and coding for the given control system

CO3: Write PLC Programming Code for specific control Applications

CO4: Apply Knowledge of Distributed control in Industrial Process Automation.

Course Contents:

Transfer Functions of Discrete-Time Systems, State Space Modeling and Realization, Stability Analysis Using Frequency Domain Methods, Linearization of Nonlinear Models, Observer-Based Controller Design Using the Pole Placement Technique PID Controller Design, PID Controller Tuning Algorithms and Simulations, Automatic Tuning of PID Controller. Simulation exercises for Speed and Torque Control of DC Motors using PID Controller. Various PWM, frequency & power control techniques. Programmable Logic Controller, SCADA, Distributed Control Systems.

Minor-IV: Industrial Internet of Things (IIoT) System design and Applications-

Prerequisites: (a) Computer Communication Network, (b) Basics of Cryptography

Course Outcomes: After successful completion of the course, students will be able to

CO1: identify the IoT components

CO2: understand & differentiate IoT architectures

CO3: select IoT protocols & Platforms

CO4: design and develop an IoT for a specific Industrial Application.



S. Chaudhary



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Autonomous Institute Affiliated to University of Mumbai)

Department of Electronics, Electronics & Telecom. Engineering

Course Contents:

IoT Architectures, Functional blocks of IoT ecosystem, Physical design of IoT, Logical design of IoT, IoT Enabling Technologies, and Security Engineering for IoT Development. The IoT Security Lifecycle-DevSecOps, Enterprise Cloud Security Architecture, Compliance Monitoring Program for the IoT, IoT Incidence Response. IoT Security and Machine Learning. Industrial IoT Levels & Application domains, Industrial IoT business model and reference architecture.



Shandhu

IT for Business
SPIT Elective from IM Area SPJIMR

Credit	1.5	Faculty	Prof. Milind Kamat, Prof. Jyoti Jagasia, Prof. Aditi Divatia & Prof. Ashutosh Jha,	Sessions	24
				Duration	70 Min. each

By the end of the course, students will have a broad perspective on how digital technologies are impacting the business space.

I. Course Overview/Background

Information Technology (IT) is fast changing the world around us. This course will provide you with an understanding of how IT-enabled Digital Technologies are changing the business environment and how insightful executives and firms leverage Digital Technologies to create value and win competitive battles. This course also introduces business analytics for individuals, including those with no prior analytics experience. You'll understand how to interpret a real-world data set and make appropriate business strategy recommendations.

II. Learning Outcomes/Course Objectives

1. To have abroad understanding of Digital Economy and Digital Transformation
2. To have a broad experience of Business Analytics, Big Data Analytics and its Use Cases in Business
3. To have a general knowledge of the Emerging world of Bots, Robotic process automation and Chatbots and Use cases in Business
4. To have knowledge of Python, What are application programming Interfaces(APIs), How APIs can be used to Integrate Chatbots and RPA bots
5. To know the coming age of 5G and It's the business impact

III. Pedagogy and Learning Style

The classes will be a mix of lecturing and discussion with cases and in-class exercises support. Students are advised to be well prepared before every class with (if given) pre-assigned readings, including articles, reports, case studies and mini-cases. Python, RPA, Chatbot modules will be having labs

IV. Course Evaluation

The evaluation components for this course will include the following:

Components	Weightage
Quizzes (2)	20%
Group Project	50%
End Term	30%



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V. Session plan

Session No./ Faculty	Learning Objective	Topics
Theme 1: Introduce to Digital: Trends and Technologies shaping Digital revolution		
1-2 MK	Understand Technology Trends shaping the Digital Economy	What is the Digital Economy? Its Size. What technologies will shape the economy going into 2025? What is Digital Disruption? What differentiates Digital Transformation from Just Digitization and Digitalization
	Understand Different Digital Technologies driving Digital Transformation	What are Digital Technologies driving Digital Transformation? Overview of blockchain, Digital currency
Theme II: Robotic Process Automation & Chatbots: New age of conversational automation		
3 JJ	RPA Part I - What is RPA, and what is not? - Evolution of RPA Automation - RPA v/s Test Automation	- Introduction RPA, - Introduction to UiPath
4 JJ	RPA-Part 2 - How to build a simple S/W robot? - What is RPA the best fit for? - Benefits, Risks and Challenges of RPA - Automation Maturity Levels - Current RPA tools	- How to build RPA BOT? - Benefits of RPA? - Challenges w.r.t RPA
5 JJ	RPA-Part 3 - Using iRobot & Orchestrator - Some examples of RPA - Web Data Scraping Examples (2) - Use cases in Industry	Use Cases of RPA in various Industries
6 JJ	RPA-Part 4 - PDF to text - OCR to text - Text to Speech – Reusable workflow - Handwriting recognition - Example use cases in the industry	Use Cases of RPA in various Industries



Shandha

7-8 JJ	Chatbot-Part 1 & 2 • Introduction Chatbots, • Introduction to DialogFlow	• Chatbot Overview • Chatbot maturity levels • Chatbots in market • Example of building a Chatbot • Uses of Chatbots • Advantages & Challenges of Chatbots • Example chatbot development in Dialogflow
Quiz 1 10% 10 Minutes		
Theme III: Data Analytics: Different types of Analytics, Big data Analytics, Concepts and Tools		
9-10 AD	Introduction to Business Analytics Text Analytics	• Introduction to Text Analytics
11-12 AD	Hands on	• Working with tools to find insights from data
13-14 AD	Hands on	• Data interpretation and action thereof from text analytics reports
Theme IV: Python scripting , AI/ML for data science		
15-16 MK	Python for data science	• Introduction to Python for data science
17-18 AJ	Management of IT: Emerging Research areas	• Introduction to the domain of "Technology Management" from a Research Point of view • Conducting Literature Review using Journal Databases
19-20 AJ	Preparation for Research paper publishing	• Journal Article Structure and writing the manuscript for different research types. • Journal Article Submission and Publication Process in reputed outlets.
Quiz 2 10% 10 Minutes		
Theme VI: Information management Research		
21-22 MK	Python for data science Panda's library	• Exploratory data analysis • Data wrangling
23-24 MK	No code, Low code Tools for AI/ML	• KNIME tool for AI/ML,
End Term Exam: 30%		
Group Project 50% Submission due TBD		

VII. Other Notes:

1. **Attendance** in the course is to be governed by program rules.
2. Participants are expected to have a laptop with at least 8GB of memory, windows running on the laptop.
3. Group Project work will be based on Text analytics & will be coordinated by prof. Milind Kamat
5. Please arrive **on** time. Attendance will not be given if you arrive late (10 minutes' post-class begins), although you may be allowed to attend the class.



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6. Contact details of Prof. Milind Kamat will be provided to all students for any clarification on group work. He will be available at a specific time at his SPJIMR office for a meeting

VII. Group project

- Each academic group will work on text analytics project as detailed below.
- Please generate 2-page proposal first as soon as you finish text analytics sessions and get it approved from prof. Milind kamat
- Proposal should have following
 - 5 consumer Brands in India (please ensure NO groups select the same brands: coco of class to ensure this)
 - Collect 3 months of all brand posts and create dataset for analytics as follows
 - Date wise posts
 - For each post, video, text, gif and message
 - User engagement generated e.g. total reactions (likes), Comments, shares
 - Then carry out Text analytics on this data and generate following
 - Word cloud
 - Sentiment analysis of comments
 - Topic modelling for comments
 - Submission for evaluation
 - Dataset you created
 - A short report of your Text Analytics
 - Introduction
 - Data collection method for 5 consumer brands
 - Your method for text analytics
 - Your conclusion based on word cloud, topics and sentiment analysis of comment
 - You are free to use any open source tools as it may be convenient to you to carry out this project work



Shandhu

Subject	Sessions	Hours/Week	Hours/ semester (14 weeks)	Credit
User Interface Design	Lecture	1	42 Hours/Sem.	3Credits
	Practical	4		
	Tutorial			
	Studio Work			

Course Objectives:-

- Demonstrate skills for low-fidelity prototyping.
- To develop enhanced visual sense for high fidelity wireframes.
- To create digital visual user interface for applications on mobile, web and laptop.

Course Content:

S. No.	Hours	Content	Detail	Lecture (Hours)	Practical (Hours)
1	3	UI Design Methodology	• Introduction to User Interface Design • Scope and relevance of UI Design • Communicating the UI • Introduction to Experience Design • Understanding process and workflow	1	2
2	6	User Research and Personas	• Understanding usability • Understanding user experience and cognition • Human - machine interaction • Stages of user's journey • Creating Personas	2	4
3	6	Introduction to User Experience	• Overview of UX Design • UX Design Process • Early design iterations • Organizing & labeling content • Visual perception and Memory • How people perceive information • How people act in the world • Environment and context Scenarios • Layout • Develop Low-fidelity wireframes • Sitemap & flow map	2	4
4	3	Gamification	• Understanding Gamification in context to design • Increasing and enhancing user engagement by weaving	1	2



Shandhar

			suitable fun features		
5	9	High Fidelity UI Design (using current software like Adobe XD, Sketch, Figma etc.)	• Mood Boards And Resources For Hi Fidelity UI Design • Why Do You Need A 12 Column Grid • How To Create A 12 Column Grid • Working With Colors • Use Web Safe Fonts: iOS Or Android Specific Fonts • How To Use Character Styles • Creating Realistic Buttons In Adobe XD With Paste Properties • How To Draw Your Own Custom Icons • How To Make A Gradient And Blur • The Pros And Cons For Working With Images • Working With Illustrator, Photoshop And Indesign In Adobe XD • How To Mask • Production Video: Adding Text Fields And Buttons • How To Make A Popup Modal With A Blurred Background • Creating 12 Column Card Dashboard Using The Repeat Grid Tool • How To Make A Mobile Phone Mockup With Adobe XD And Photoshop • Hidden Features For Adobe XD Repeat Grid • Advanced Prototyping Using Buttons And Dropdown Menus • One More Prototype Trick And Faking Depth Of Field For Mockup • How To Export Images And Code From Adobe XD For Developers • Shortcuts And Cheat Sheet	2	6
6	6	Prototyping and Heuristic Evaluation	• Visual Design • UI Semiotics • Usability inspection • Develop High-fidelity wireframes • Pre-testing	2	4
7	6	User Interface for Various platforms	• Understanding difference in interface of - Mobile Devices	2	4



Shandhuji

			- Tablet - Web - Laptop - Software		
8	2	iOS vs Android	- Understanding interface and usability difference between iOS vs Android - Post-testing	2	1

INDICATIVE ASSIGNMENT BRIEF AND EVALUATION MATRIX

Mid- Assessment : 30 Marks Brief: Select an area / existing application to work on (may be given by faculty or selected by students eg. Food delivery, women safety, fitness and health etc) Study the navigation pathway and pain points of the app chosen. Suggest alternatives to address any issues in navigation and pain points.	End – Term Evaluation (Exam/Jury): 40 Marks Brief: Find a local charity / NGO and create an intuitive interface for them by designing / redesigning a donation portal for them. Students are expected to use gamification in order to enhance the overall experience in order to make people derive pleasure and sense of accomplishment post donations. Students may also do post user testing of the interface and improve the usability to create a final and functional design.
End Assessment : 30 Marks Brief : Re-design the app selected in Assignment 1 by adding / deleting tabs / buttons, changing navigation in order to minimize cognitive load (Cognitive load refers to the amount of brain power required to navigate the over information provided by the app). Students are expected to develop user personas, sketches of initial ideas, sitemap/journey map, wireframes and working prototype.	
EVALUATION PARAMETER / CRITERIA • Critical and Design Thinking ability-40% • Process of Ideation-30% • Presentation & Communication-30%	
Marks 30+ 30	Marks :40
Mid-Term/ Mid-Module Assessment, End-Term/ End-Module Assessment	End Term Evaluation (Exam/Jury)

REFERENCE MATERIAL/BOOKS:

- Allanwood, G., Beare, P., 2014. Basics Interactive Design: User Experience Design. China : Bloomsbury Publishing
- Wood, D., 2014. Basics Interactive Design: Interface Design. China : Bloomsbury Publishing
- Grant, W., 2018. 101 UX Principles. Packt Publishing Limited
- Norman, D. *The Design of Everyday Things*: Basic Books Publishing
- Rogers, Y., Sharp, H., Preece, J. *Interaction Design: Beyond Human-Computer Interaction*: Wiley Publishing
- Tufte, E.R. *Envisioning Information*: Graphics Press
- Lazar, J., Feng, J.H., Hochheiser, H. *Research Methods in Human-Computer Interaction*: Wiley Publishing

INFRASTRUCTURE REQUIREMENT:

Physical space

- Computer Lab/ facilities for using own Systems / Laptops with AdobeXD Installed on all Systems/Laptops
- Digital projector / overhead projector and computer for illustrated lectures
- Classroom and Digital Studio with soft boards



Signature

EQUIPMENT/TOOLS/MATERIAL

- Well-equipped lab with Software Application installed(Adobe XD,Adobe Photoshop)



Shandhu

Subject	Sessions	Hours/ Week	Hours/ Semester (14 weeks)*	Credits	Subject Type
Data Representation and Analysis	Lecture	2	42	3	Major Core
	Practical	2			
	Studio Work				
	Studio Practice				

**Teaching sessions to be planned for 14 weeks, excluding end term examination week.*

SUBJECT AIM

To understand the human-centered approach to problem solving, and employ the concepts of sustainability and systems-level change for addressing critical social needs.

LEARNING OUTCOMES OF THE SUBJECT

Knowledge

- To understand design as a problem solving tool.
- To understand the cognitive and instructional aspects in defining an effective workflow
- To visualize, design and interpret workflow and information architecture
- To develop intuitive digital content strategy.

Skill

- To employ design as a problem solving tool
- To develop strategies and innovative approaches for addressing social needs

Synthesis/Outcome

- Empathize and appreciate 'human' as an integral element of a society rather than the 'products' and 'market'.

CONTENT

S No.	Hours	Content	Detail	Lecture	Practical
1	4	Design as a problem-solving tool and process	• Understanding the Design Process • Define the problem	3	1



S. Chaudhary

			• Empathy map • Journey map • Re-define • Cynefin Framework • Drivers of change • Reframing • Share & connect • Evolve ideas		
2	4	UX Research techniques	• Interviewing • Field Visits • Card sorting • Types of card sorting Open card sorting, Closed card sorting, • Affinity Mapping	2	2
3	6	Introduction to Brainstorming	• Importance of brainstorming • Brainstorming as an idea generation tool. • Limitations of brainstorming • Brainstorming Techniques for UI / UX	2	4
4	6	Implications for Design	• Setting out your ideas and intentions in a well-presented clear manner • Scenario mapping • Identifying design opportunities • Communicating Ideas to Stakeholders	2	4
5	4	Information Architecture (IA)	• What is IA • importance of IA • IA as foundation for UI/UX • Models in Information Architecture	2	2
	8	Designing an IA	• Identifying brand ethos • Brainstorming • Assessment of content requirement • Selecting appropriate model in IA	4	4



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			- Information Ecosystems - Labelling, Navigation and Search contents - Creation of IA through a IA open source software		
6	6	Social enterprise business plan	Project based	2	3
7	4	Marketing toolkits and tips	Project based	1	3

LEARNING AND TEACHING METHODS

- Lectures and/or media presentations explaining theory and issues specific to the topic
- Demonstrations of materials, techniques and processes
- Individual critiques with each student

NATURE OF ASSIGNMENTS

The students learn to understand and employ the concepts of sustainability and systems-level change for addressing critical social needs. They learn to recognise the social problems and develop a business plan to achieve a social change by employing entrepreneurial principles, processes and operations.

EVALUATION CRITERIA

- Development of concepts that are thoughtful, original and creative
- Development of technical digital skills and techniques
- Expert execution, technique and completion of projects
- Effective visual and verbal presentation skills
- Active and verbal participation in class activities and critique
- Attendance with necessary materials and assignment preparation
- Personal challenge and effort in project development
- Deadline compliance

EVALUATION MATRIX & INDICATIVE ASSIGNMENT BRIEFS

Mid Term/ Mid Module Assessment	End Term/ End Module Assessment	End Term Jury Evaluation	Total
30	30	40	100
Students will present a challenge/problem (wicked problem) and will be asked to solve the challenge using mapping methods and tools.	They learn to recognise the social problems and develop a business plan.	The students complete the project concepts of systems-level change and learning from social problems and develop a business plan to achieve a social change by employing entrepreneurial principles, pro-	



Shandhar

		cesses and operations.	
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REFERENCE MATERIAL FOR FACULTIES

Books

-
- Curedale, R., (2016). Affinity Diagrams: The tool to tame complexity
- Szabo, P., User Experience Mapping: Enhance UX with User Story Map, Journey Map and Diagrams
- Angeli Sachs et. Al, Social Design: Participation and Empowerment.
- Information Visualization- perception for Design, Colin Ware, e-book
- Visual Thinking for Design, Colin Ware, e-book
- Field Guide of Human Centred Design : IDEO
- Victor Papanek, Design for the Real World: Human Ecology and Social Change.
- Lucy Kimbell & Joe Julier, The Social Design methods Menu
- Cheryl Heller, The Intergalactic Design Guide- Harnessing the Creative Potential of Social Design.
- Tim Brown & Jocelyn Wyatt, Design Thinking for Social Innovation.
- Shea, A., Designing for Social Change

E-learning

REFERENCE MATERIAL FOR STUDENTS

Books

- Field Guide of Human Centred Design : IDEO
- Tim Brown & Jocelyn Wyatt, Design Thinking for Social Innovation.

E-learning

INFRASTRUCTURE REQUIREMENT

- Latest Mac/Window Desktop/Laptop of sufficient specs to run the latest multimedia software
- Multimedia digital display



Shandha

Subject	Sessions	Hours/Week	Hours/ semester (14 weeks)	Credit
Introduction to Visual Design	Lecture	2	42 hrs/sem	3
	Practical	1		
	Tutorial			
	Studio Work			

Course objective:

- To understand the fundamentals of visual design to create visual content which is appealing and engaging for an audience.

Learning Outcomes of the subject

Knowledge

- To understand the basic elements of visual perception
- Understanding the scope of visual communication
- Will be able to apply graphic design principles in the ideation, development, and production of visual messages.
- Apply structured design process for data exploration, analysis, and storytelling

Skill

- Apply fundamentals of visual design in creating different types of content in the realm of visual communication.
- Arrangement of visual elements for better communication and understanding

Synthesis/Outcome

- To incorporate elements and principles of design to communicate and express concepts/ideas

Course Content:



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S No.	Hours	Content	Detail	Lecture	Practical
1	3	Introduction to Visual Design	• What is Visual Design • Evolution of Visual Language - from history to contemporary	3	
2	10	Fundamentals of Visual communication	• Elements of design - understanding the building blocks of design - Point, Line, Plane, Shape, Texture, Colour, Pattern, Form • Principles of Design and Gestalt principles / Visual Design principles - Understanding & articulation of aesthetic principles - Balance, Rhythm, Harmony, Order and Movement, • Placement, Size and Proportion, Alignment, Orientation, Consistency, Hierarchy and White space	6	4
3	5	Introduction to Information Design	• Information Design Principles, Graphics Design • Fundamentals • History, Design, and Redesign of Information • Visual data exploration methods •	3	2
4	4	Storytelling, Storyboards, and Sketching	Importance of storytelling. Importance of Storyboard. Importance of • Visualization • Memorability • Empathy • Engagement •	3	1
5	6	Color Psychology, Typography	• What is Color - Properties, RGB/CMYK Model • Introduction to Typography - evolution of digital fonts • Anatomy of Type • Text Classification	4	2

INDICATIVE ASSIGNMENT BRIEF AND EVALUATION MATRIX

Mid- Assessment : 30 Marks Brief: Semantic experiments with different elements of design and to depict meanings Or Choose any visual media and research on the Infographics and their application in text, icons, images, color etc	End – Term Evaluation (Exam/Jury): 40 Marks Brief: Design/Redesign collaterals for a brand in Adobe Illustrator Or Brief : Choose a topic like Gender Pay Gap, Changing Indian diet etc. and collect data on the topic chosen (Primary or secondary source) and Create an impactful info graphic that tells in-depth stories and educates customers
End Assessment : 30 Marks Brief : Refer to existing data set from current Newspapers / websites etc. & convert it into meaningful Info graphic using the tools learnt in the class	
EVALUATION PARAMETER / CRITERIA	
Marks 30+ 30	Marks :40
Mid-Term/ Mid-Module Assessment, End-Term/ End-Module Assessment	End Term Evaluation (Exam/Jury)

REFERENCE MATERIAL

Books

- George-Palilonis, J., 2017. A practical guide to graphics reporting. 2nd Ed. New York: Routledge
- Meirelles, I., 2013. Design for information. Beverly: Rockport
- Katz, J., 2012. Designing information. Hoboken: Wiley
- Tufte, E.R., 2010. Envisioning information. Cheshire: Graphics Press
- Kim, B., 2008. Information design workbook. Beverly: Rockport
- Knaflic, C.N., 2015. Storytelling with data. Hoboken: John Wiley & Sons
- Cairo, A., 2013. The Functional Art. Berkeley: New Riders
- Smiciklas, M., 2012. The power of Infographics. Indianapolis: Que Pub
- Krum, R., 2013. Cool Infographics. Indianapolis: John Wiley & Sons
- Shaoqiang, W. ed., 2018. Playful data. Barcelona: Promopress
- Nussbaumer-Knaflic, C., 2015. Storytelling with Data
- Ambrose & G. Harris P. (2009) The Fundamentals of Graphic Design, AVA



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- Lupton E. & Philips J. C. (2008) Graphic Design The New Basics, Second Edition, Princeton Architectural Press
- Ambrose G. (2006) The Fundamentals of Typography: Second Edition. AVA
- Lupton E. and Phillips J.C. (2010) Graphic Design-The New Basics. Princeton Architectural Press

INFRASTRUCTURE REQUIREMENT

- Latest Mac/Window Desktop/Laptop of sufficient specs to run the latest multimedia software
- Multimedia digital display

EQUIPMENT/TOOLS/MATERIAL

- Software: latest version of Adobe Illustrator



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